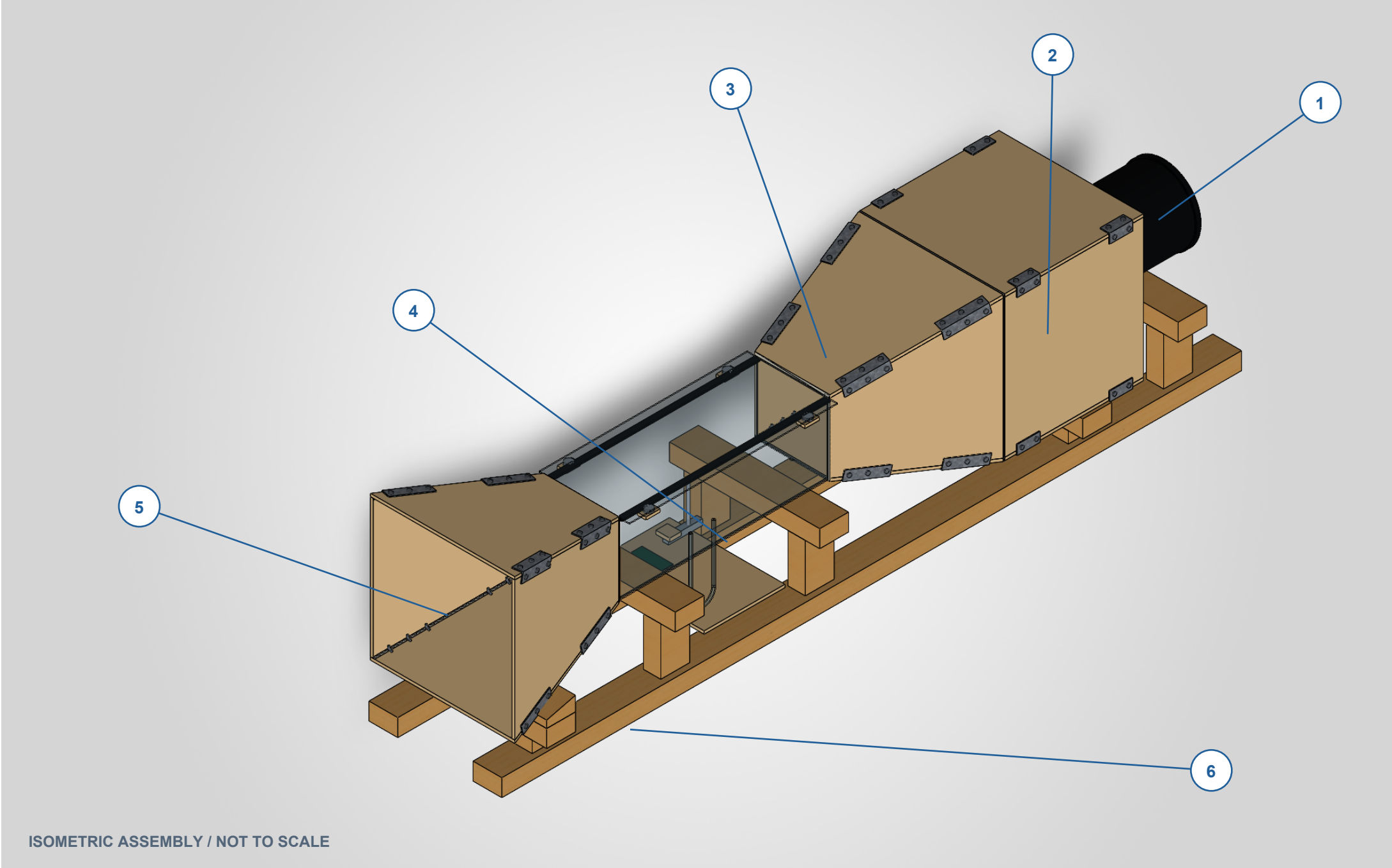


WIND TUNNEL / ENGINEERING DRAWING SET

A00 | REV D | 2026-10-01 | ASSEMBLY OVERVIEW + 50 INDIVIDUAL PART DRAWINGS



PRINCIPAL DIMENSIONS

Overall base length: 64.000 in (assumed)
Overall width: 13.000 in
Overall height: 16.660 in
Fan envelope: 6 in nominal / 8 in long
Inlet: 12 x 12 in clear / 12 in long
Contraction: 12 to 6 in / 12 in long
Test passage: 6 x 6 in / 18 in long
Diffuser: 6 to 12 in / 12 in long
Three section gaps: 0.125 in each
Test-module transverse shift: 0.125 in
Plywood: 0.250 in radial thickness
Acrylic: 0.093 in thickness

LABELED COMPONENTS

- 1 Fan: P20-P22
- 2 Inlet: P01-P04; conditioning P17-P19
- 3 Contraction: P05-P08
- 4 Test section / lid: P09-P12, P23-P25
- 5 Diffuser: P13-P16; saddle P43
- 6 Base / supports: P26-P31

234 INSTANCES / 49 INSTALLED PART TYPES

P39: separate measured reference, quantity 0.

DRAWING GUIDE

P01-P50 each have four SolidWorks model-linked views: isometric, side, plan and end. View scales vary for legibility. Overall dimension lines and feature-detail tables describe the modeled solids. Use the PDF bookmarks or the website part selector to find a sheet.

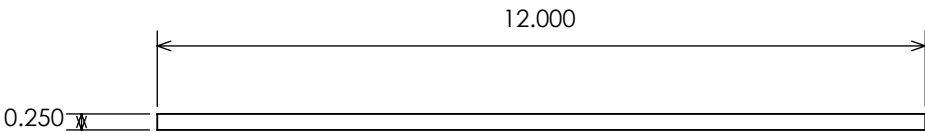
Additional groups: instrumentation P32-P36; longitudinal inside corner braces P37-P38; measured panel P39; retainers, screws, U-tube and saddle P40-P43; separate caulk parts P44-P50.

Dimension basis: project dimensions, BOM and build photos, supplemented by nominal fit assumptions. Precise as-built gaps, bracket patterns and purchased-component internals were not measured. See Parts_register.csv for each part's basis.

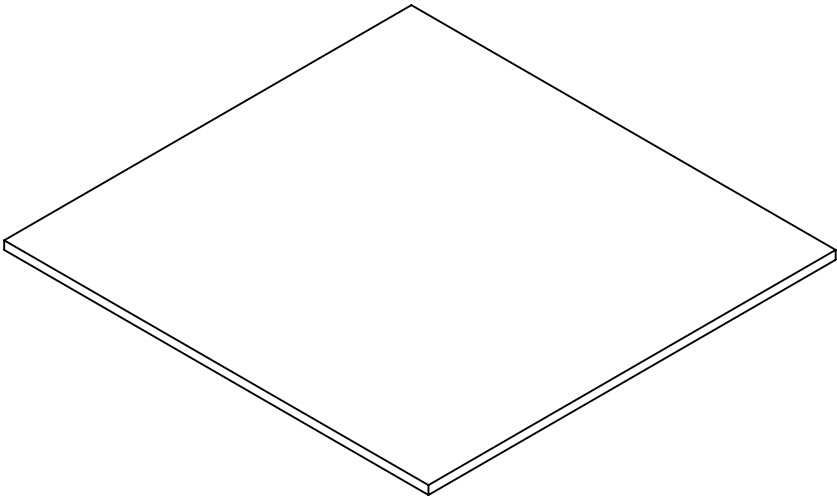
WIND TUNNEL / P01 / INLET TOP

WT-P01 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

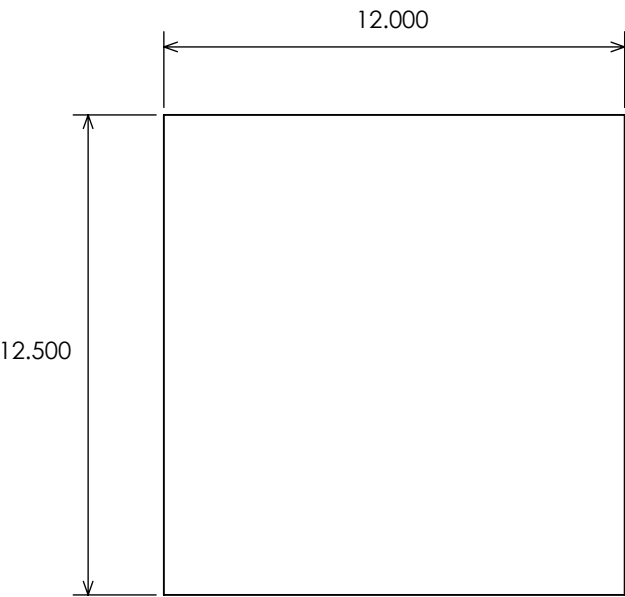
SIDE / X-Z | 1:3



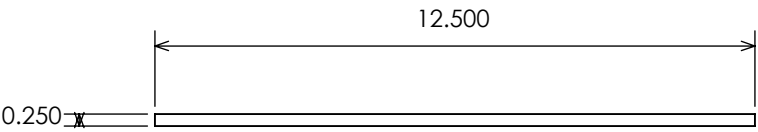
ISOMETRIC | 1:4



PLAN / X-Y | 1:5



END / Y-Z | 1:4



PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Panel width: 12.500
- Wall thickness: 0.250
- Clear inlet: 12.000 square
- Panel is a rectangular solid.
- Butt joints follow CAD assembly.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

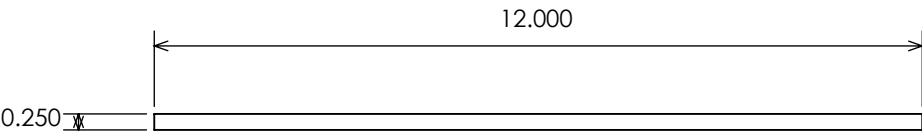
ASSEMBLY QTY: 1 | 01_Inlet_Top.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

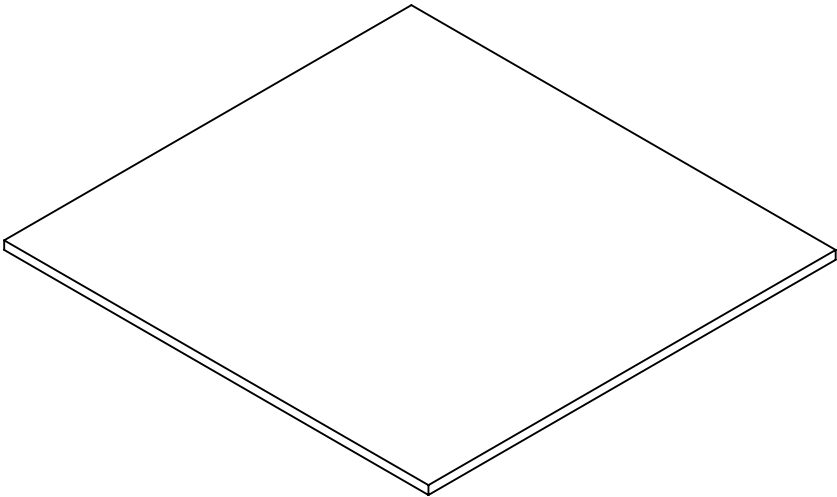
WIND TUNNEL / P02 / INLET BOTTOM

WT-P02 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

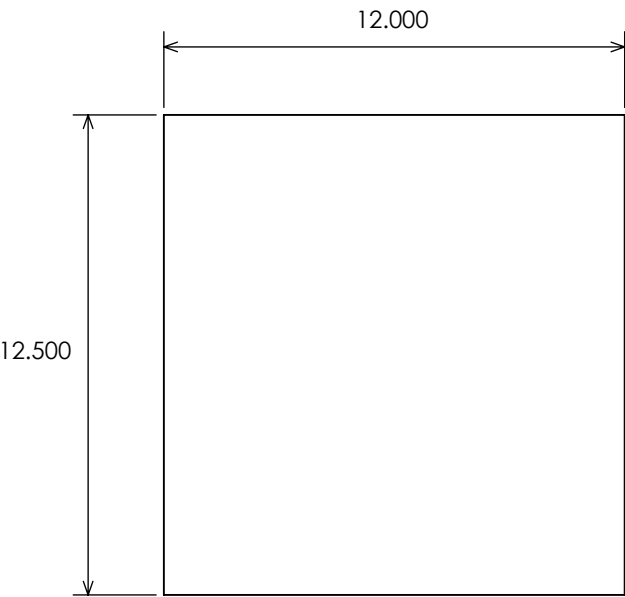
SIDE / X-Z | 1:3



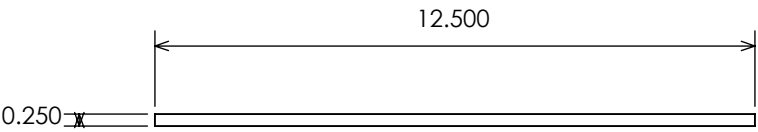
ISOMETRIC | 1:4



PLAN / X-Y | 1:5



END / Y-Z | 1:4



PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Panel width: 12.500
- Wall thickness: 0.250
- Clear inlet: 12.000 square
- Panel is a rectangular solid.
- Butt joints follow CAD assembly.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

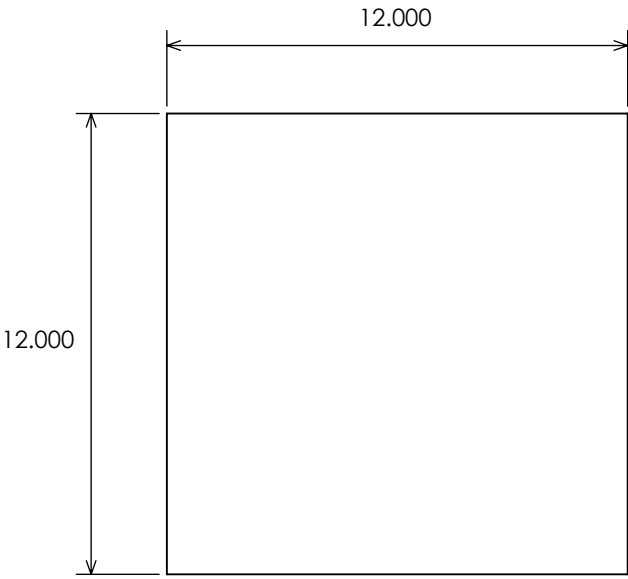
ASSEMBLY QTY: 1 | 01_Inlet_Bottom.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

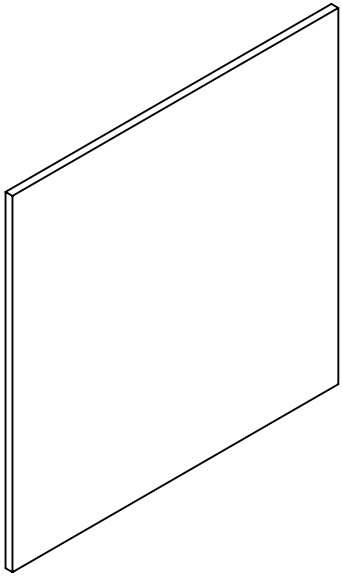
WIND TUNNEL / P03 / INLET RIGHT

WT-P03 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:5



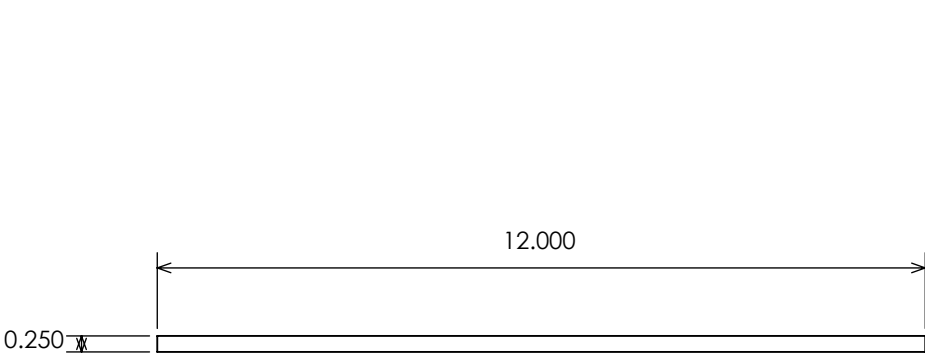
ISOMETRIC | 1:5



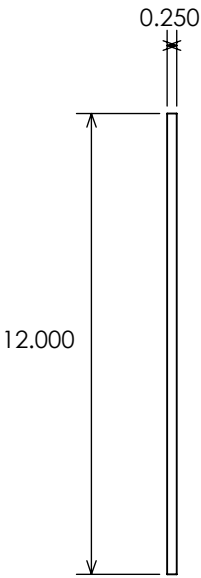
PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Panel width: 12.000
- Wall thickness: 0.250
- Clear inlet: 12.000 square
- Panel is a rectangular solid.
- Butt joints follow CAD assembly.

PLAN / X-Y | 1:3



END / Y-Z | 1:5



MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

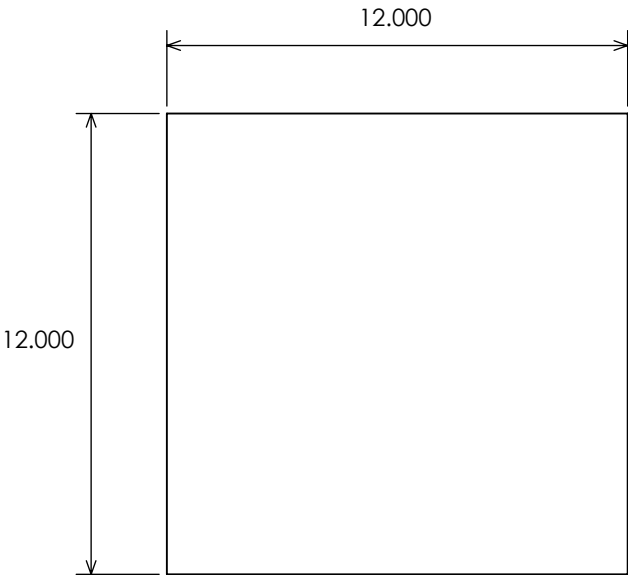
ASSEMBLY QTY: 1 | 01_Inlet_Right.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

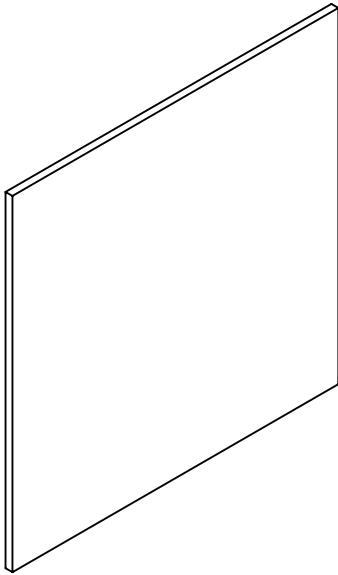
WIND TUNNEL / P04 / INLET LEFT

WT-P04 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

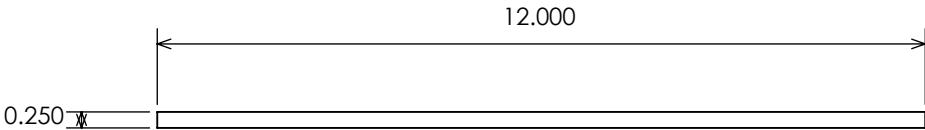
SIDE / X-Z | 1:5



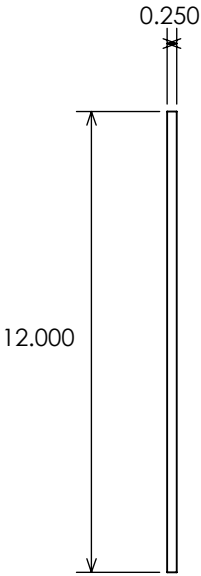
ISOMETRIC | 1:5



PLAN / X-Y | 1:3



END / Y-Z | 1:5



PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Panel width: 12.000
- Wall thickness: 0.250
- Clear inlet: 12.000 square
- Panel is a rectangular solid.
- Butt joints follow CAD assembly.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

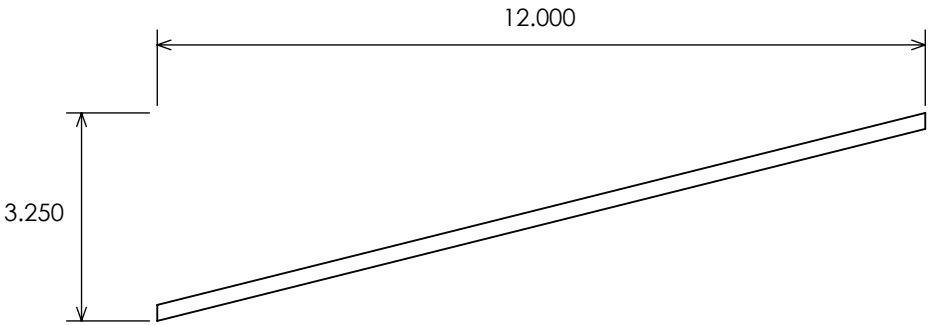
ASSEMBLY QTY: 1 | 01_Inlet_Left.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

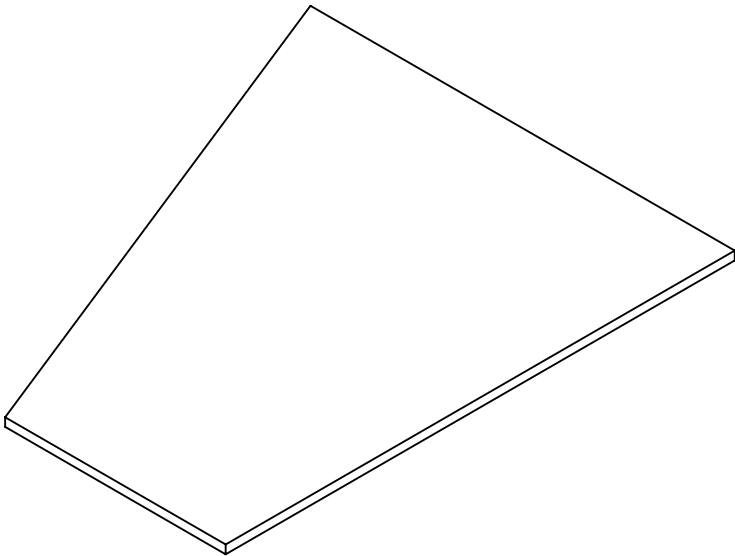
WIND TUNNEL / P05 / CONTRACTION TOP

WT-P05 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

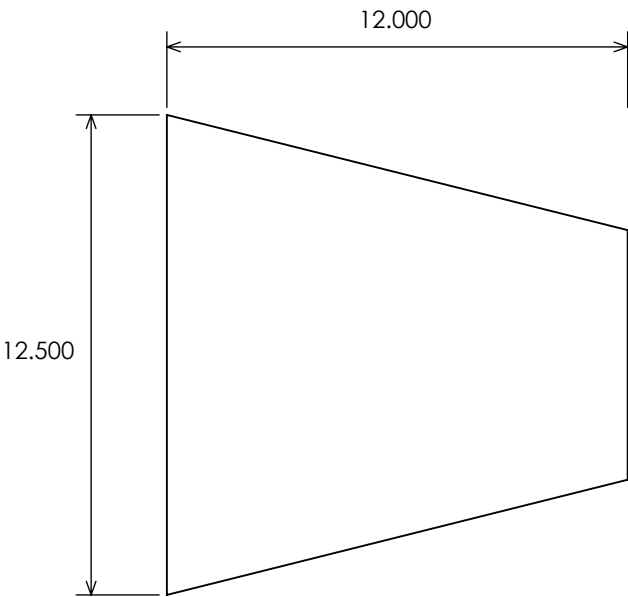
SIDE / X-Z | 1:3



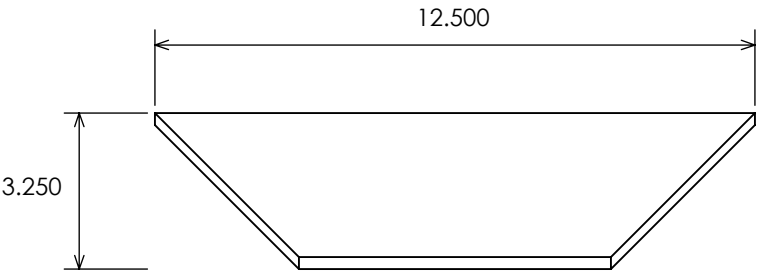
ISOMETRIC | 1:4



PLAN / X-Y | 1:5



END / Y-Z | 1:4



PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Entry panel width: 12.500
- Exit panel width: 6.500
- Radial wall thickness: 0.250
- Normal thickness: 0.242536
- Wall rise / fall: 3.000
- Wall slope: 14.0362 deg
- Centerline face length: 12.369317
- Inner edge seam: 12.727922
- Clear bore: 12.000 to 6.000 square
- Ends perpendicular to X axis.
- Lofted installed panel; not a flat pattern.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

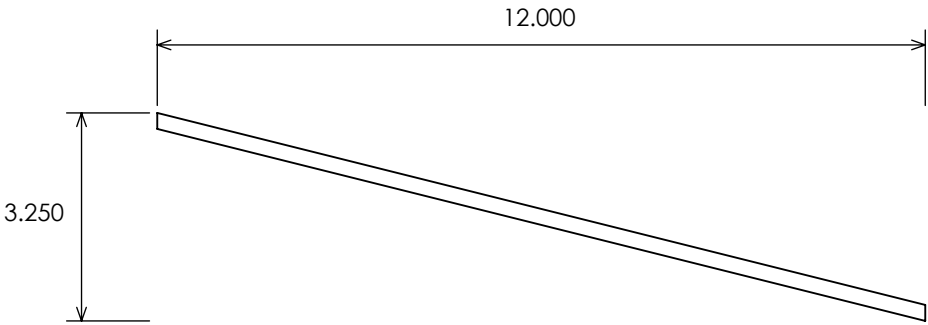
ASSEMBLY QTY: 1 | 02_Contraction_Top.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

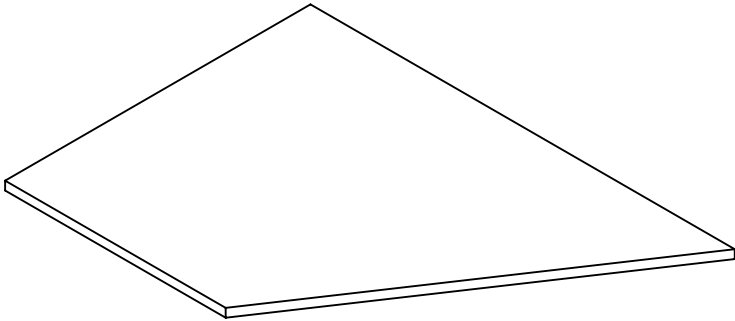
WIND TUNNEL / P06 / CONTRACTION BOTTOM

WT-P06 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

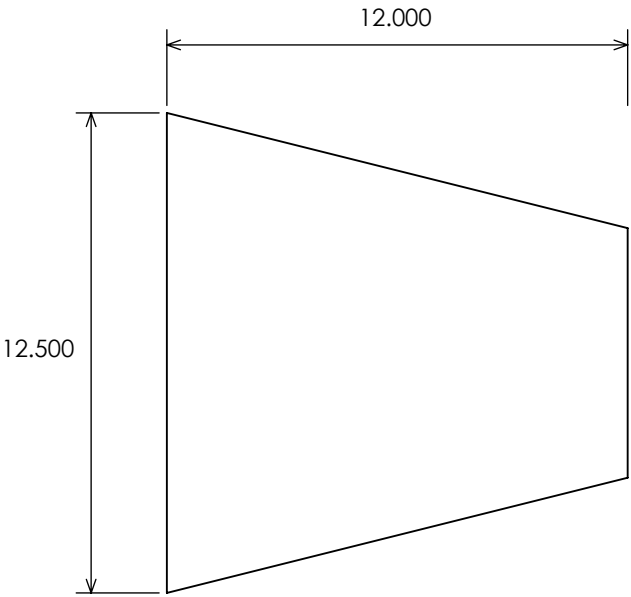
SIDE / X-Z | 1:3



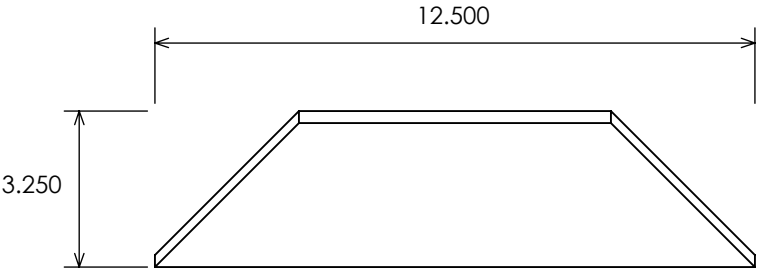
ISOMETRIC | 1:4



PLAN / X-Y | 1:5



END / Y-Z | 1:4



PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Entry panel width: 12.500
- Exit panel width: 6.500
- Radial wall thickness: 0.250
- Normal thickness: 0.242536
- Wall rise / fall: 3.000
- Wall slope: 14.0362 deg
- Centerline face length: 12.369317
- Inner edge seam: 12.727922
- Clear bore: 12.000 to 6.000 square
- Ends perpendicular to X axis.
- Lofted installed panel; not a flat pattern.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

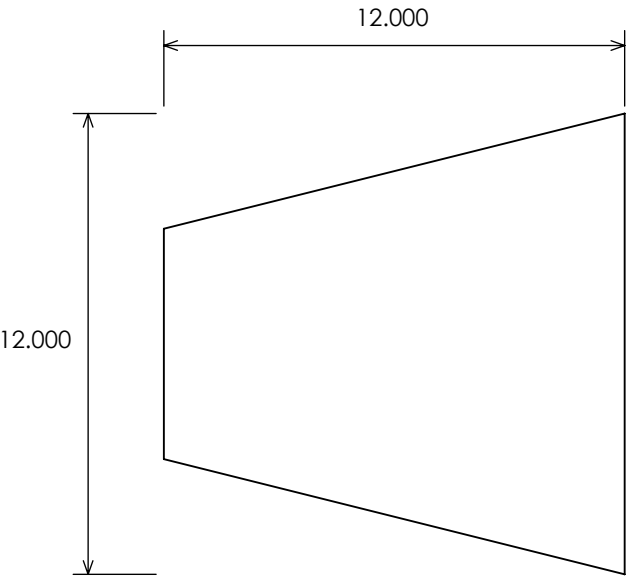
ASSEMBLY QTY: 1 | 02_Contraction_Bottom.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

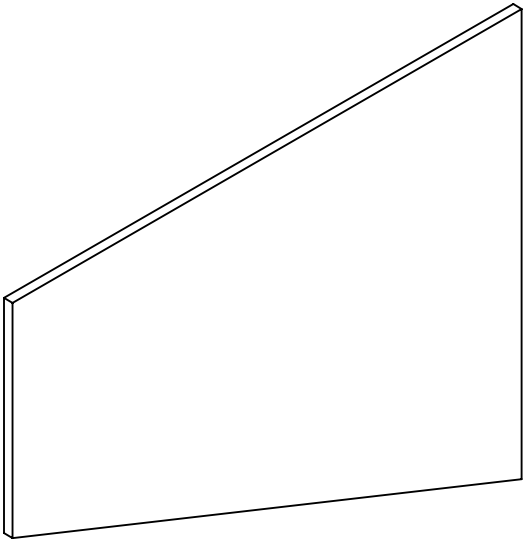
WIND TUNNEL / P07 / CONTRACTION RIGHT

WT-P07 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:5



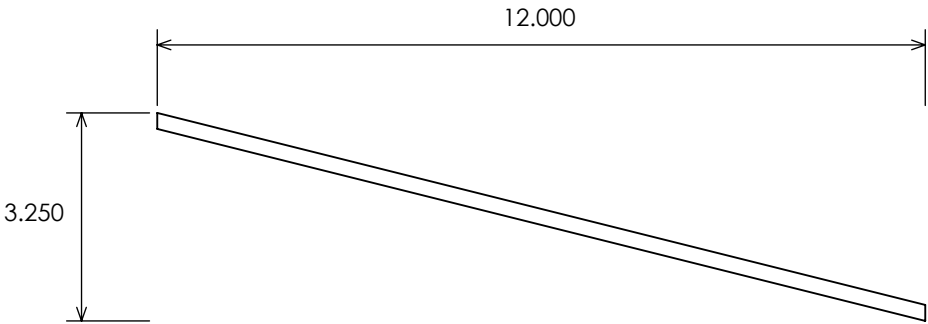
ISOMETRIC | 1:4



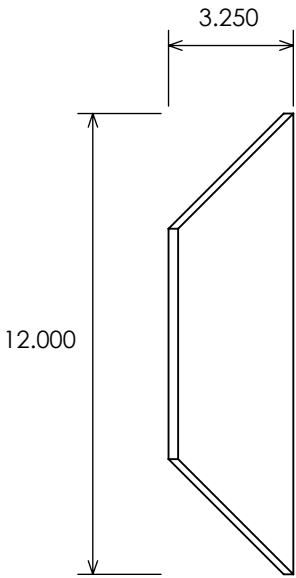
PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Entry panel width: 12.000
- Exit panel width: 6.000
- Radial wall thickness: 0.250
- Normal thickness: 0.242536
- Wall rise / fall: 3.000
- Wall slope: 14.0362 deg
- Centerline face length: 12.369317
- Inner edge seam: 12.727922
- Clear bore: 12.000 to 6.000 square
- Ends perpendicular to X axis.
- Lofted installed panel; not a flat pattern.

PLAN / X-Y | 1:3



END / Y-Z | 1:5



MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

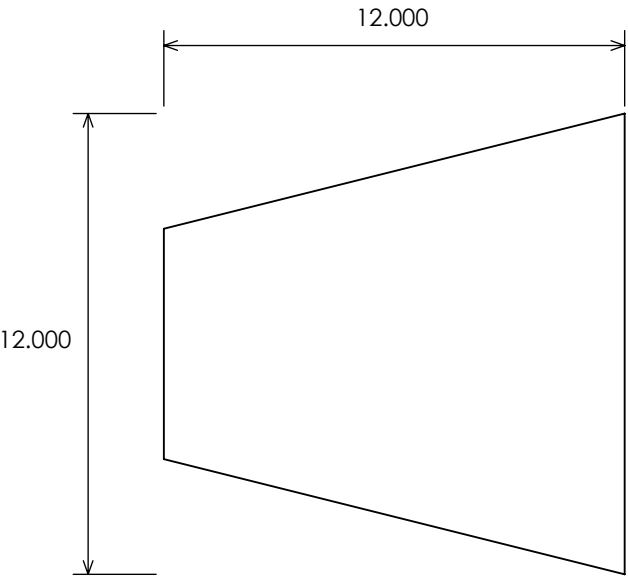
ASSEMBLY QTY: 1 | 02_Contraction_Right.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

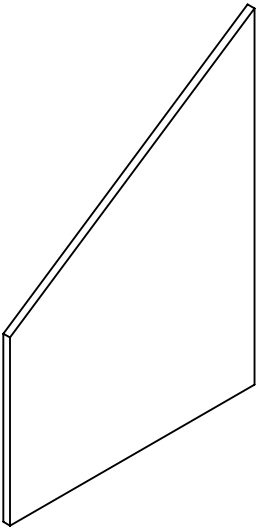
WIND TUNNEL / P08 / CONTRACTION LEFT

WT-P08 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

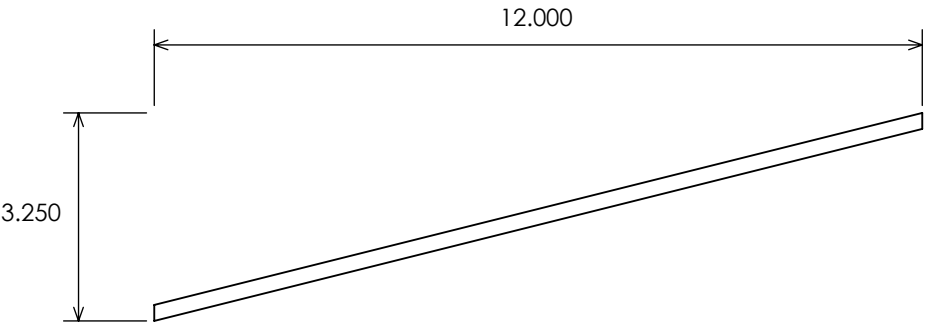
SIDE / X-Z | 1:5



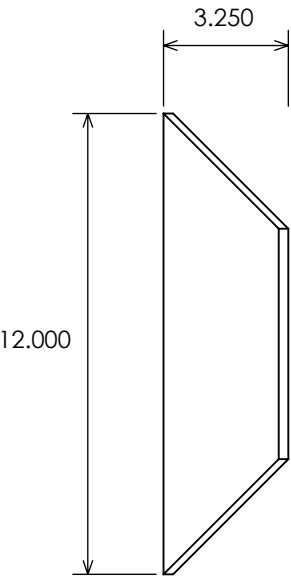
ISOMETRIC | 1:5



PLAN / X-Y | 1:3



END / Y-Z | 1:5



PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Entry panel width: 12.000
- Exit panel width: 6.000
- Radial wall thickness: 0.250
- Normal thickness: 0.242536
- Wall rise / fall: 3.000
- Wall slope: 14.0362 deg
- Centerline face length: 12.369317
- Inner edge seam: 12.727922
- Clear bore: 12.000 to 6.000 square
- Ends perpendicular to X axis.
- Lofted installed panel; not a flat pattern.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

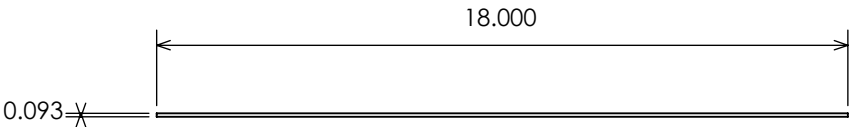
ASSEMBLY QTY: 1 | 02_Contraction_Left.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

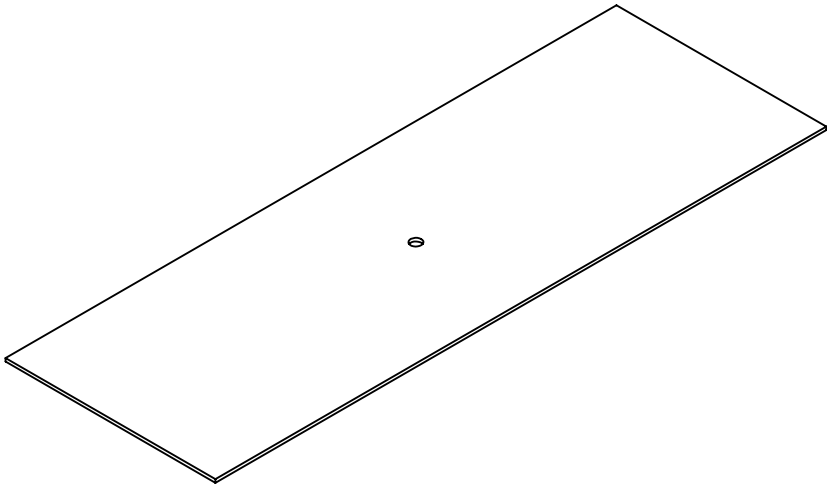
WIND TUNNEL / P09 / TEST FLOOR

WT-P09 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

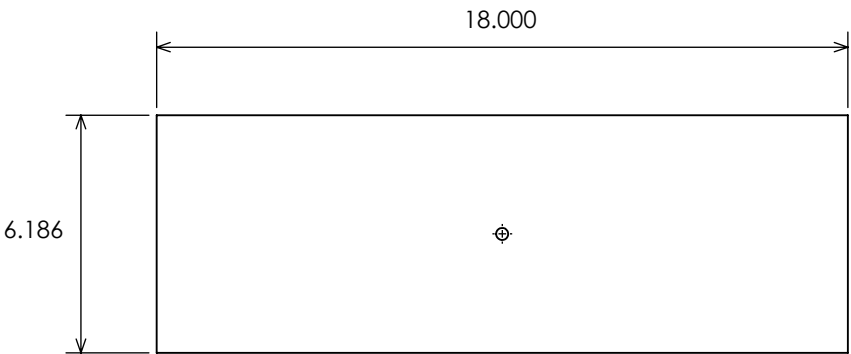
SIDE / X-Z | 1:5



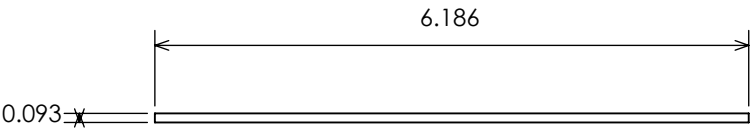
ISOMETRIC | 1:4



PLAN / X-Y | 1:5



END / Y-Z | 1:2



PART DIMENSIONS / DETAILS

Length: 18.000
Width: 6.186
Thickness: 0.093
Rod clearance: DIA 8 mm THRU
Hole center: X = 9.000
Hole center: Y = 0.000
From either long edge: 3.093
Hole placement is assumed.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Acrylic (Medium-high impact)

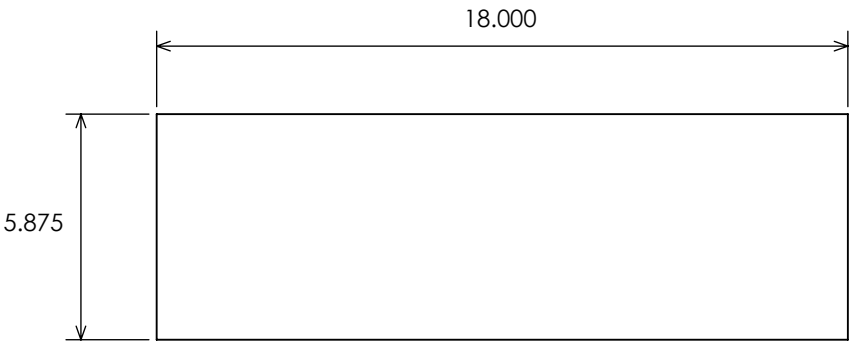
ASSEMBLY QTY: 1 | 03_Test_Floor.SLDPR

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

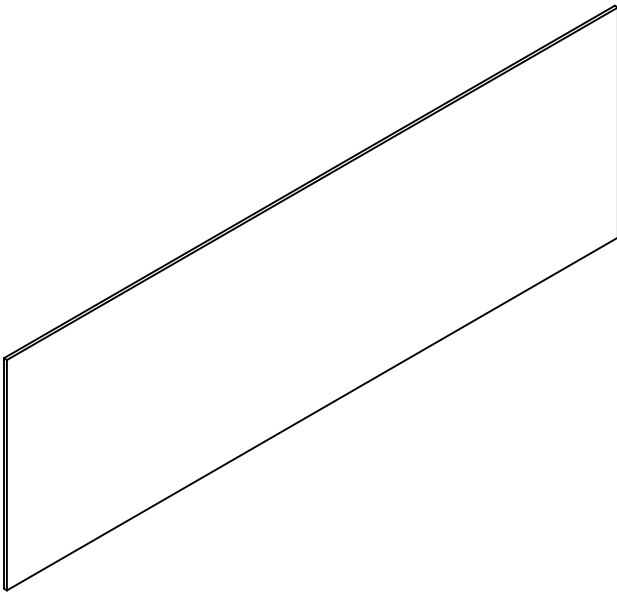
WIND TUNNEL / P10 / TEST RIGHT

WT-P10 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

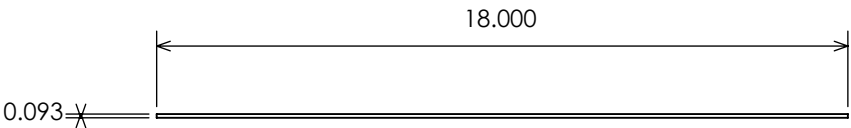
SIDE / X-Z | 1:5



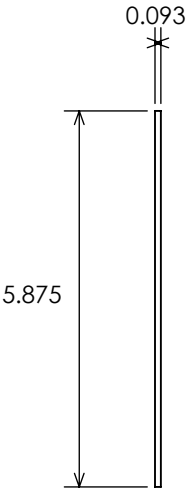
ISOMETRIC | 1:4



PLAN / X-Y | 1:5



END / Y-Z | 1:3



PART DIMENSIONS / DETAILS

Length: 18.000
Height: 5.875
Thickness: 0.093
Top edge Z = 2.875
Bottom edge Z = -3.000
Gasket allowance: 0.125

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Acrylic (Medium-high impact)

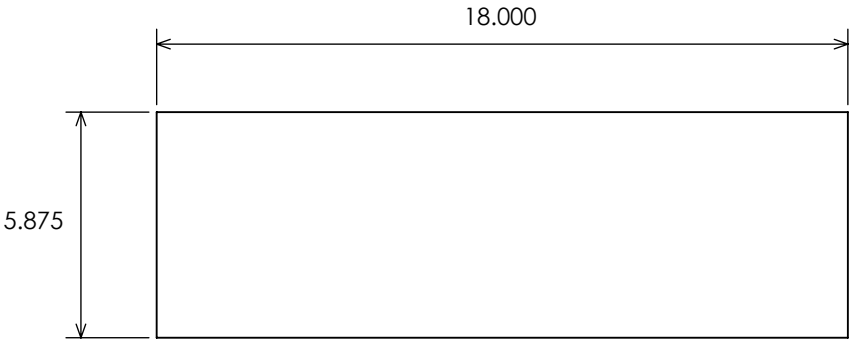
ASSEMBLY QTY: 1 | 03_Test_Right.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

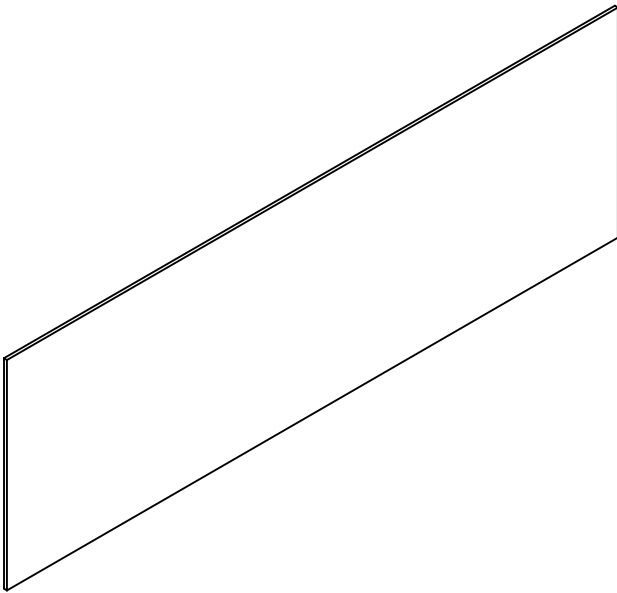
WIND TUNNEL / P11 / TEST LEFT

WT-P11 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:5



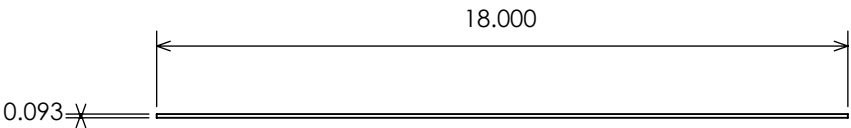
ISOMETRIC | 1:4



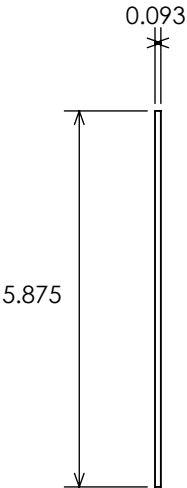
PART DIMENSIONS / DETAILS

Length: 18.000
Height: 5.875
Thickness: 0.093
Top edge Z = 2.875
Bottom edge Z = -3.000
Gasket allowance: 0.125

PLAN / X-Y | 1:5



END / Y-Z | 1:3



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Acrylic (Medium-high impact)

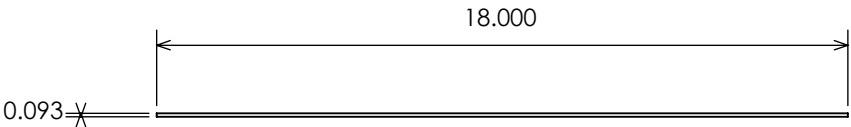
ASSEMBLY QTY: 1 | 03_Test_Left.SLDPR

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

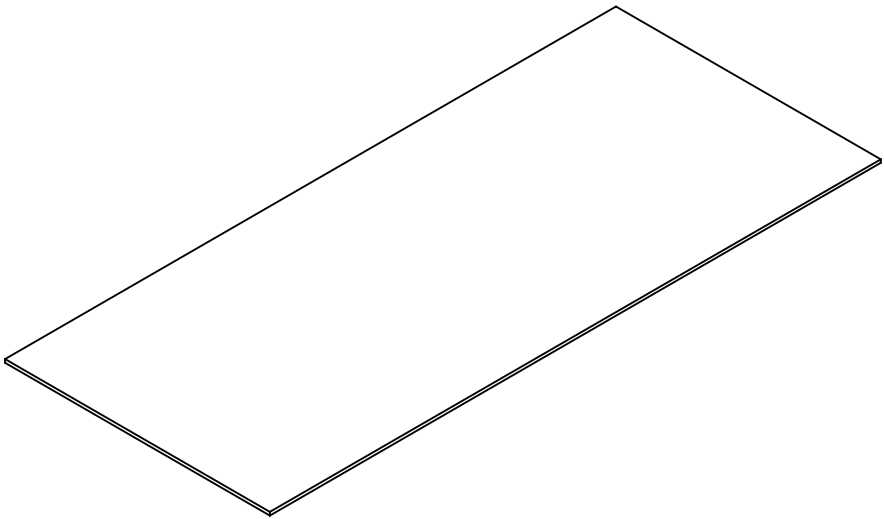
WIND TUNNEL / P12 / REMOVABLE LID

WT-P12 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

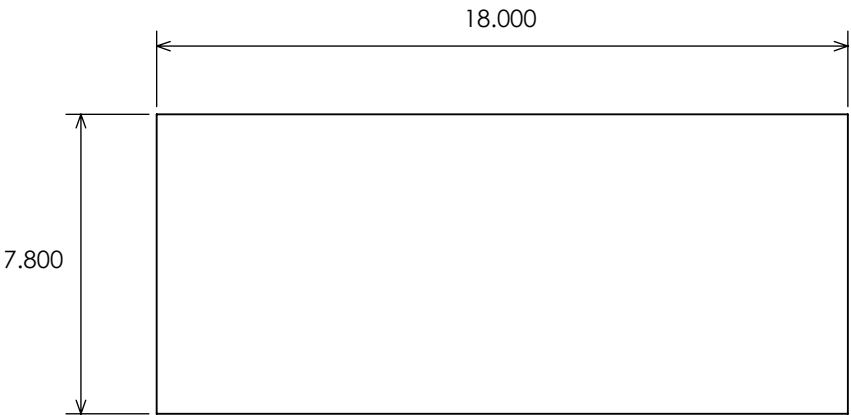
SIDE / X-Z | 1:5



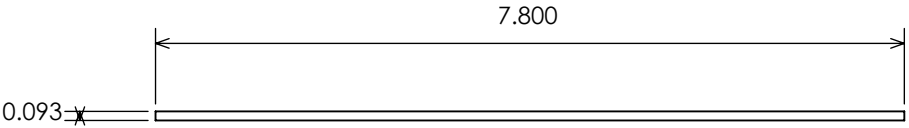
ISOMETRIC | 1:4



PLAN / X-Y | 1:5



END / Y-Z | 1:2



PART DIMENSIONS / DETAILS

- Length: 18.000
- Width: 7.800
- Thickness: 0.093
- Width increased for magnetic tabs.
- Nominal passage height: 6.000

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: Acrylic (Medium-high impact)

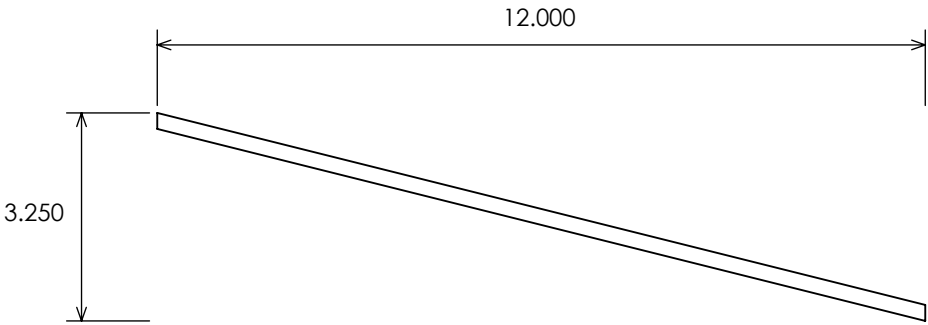
ASSEMBLY QTY: 1 | 03_Removable_Lid.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

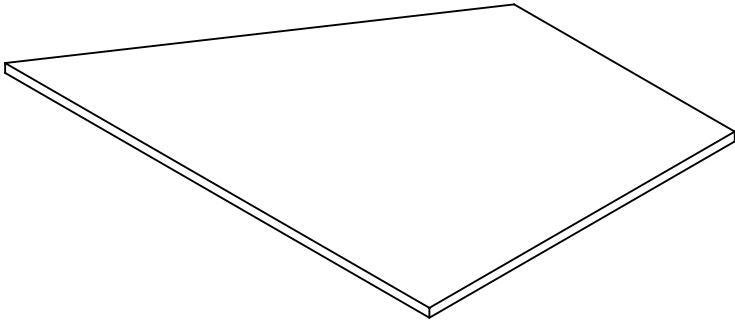
WIND TUNNEL / P13 / DIFFUSER TOP

WT-P13 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

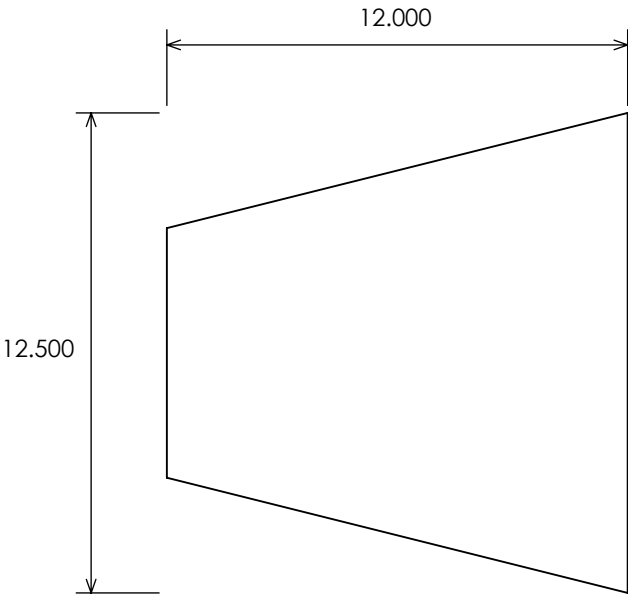
SIDE / X-Z | 1:3



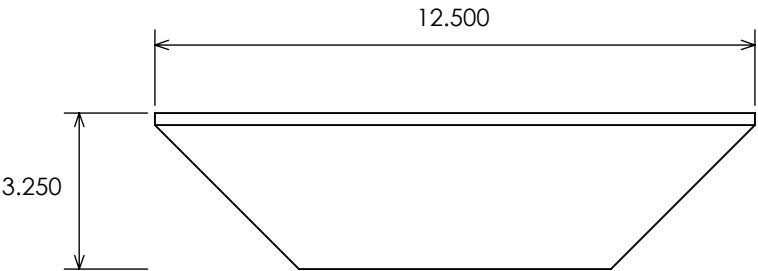
ISOMETRIC | 1:4



PLAN / X-Y | 1:5



END / Y-Z | 1:4



PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Entry panel width: 6.500
- Exit panel width: 12.500
- Radial wall thickness: 0.250
- Normal thickness: 0.242536
- Wall rise / fall: 3.000
- Wall slope: 14.0362 deg
- Centerline face length: 12.369317
- Inner edge seam: 12.727922
- Clear bore: 6.000 to 12.000 square
- Ends perpendicular to X axis.
- Lofted installed panel; not a flat pattern.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

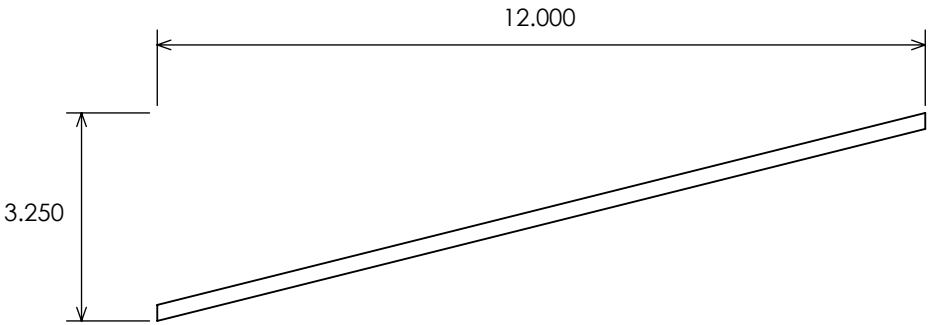
ASSEMBLY QTY: 1 | 04_Diffuser_Top.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

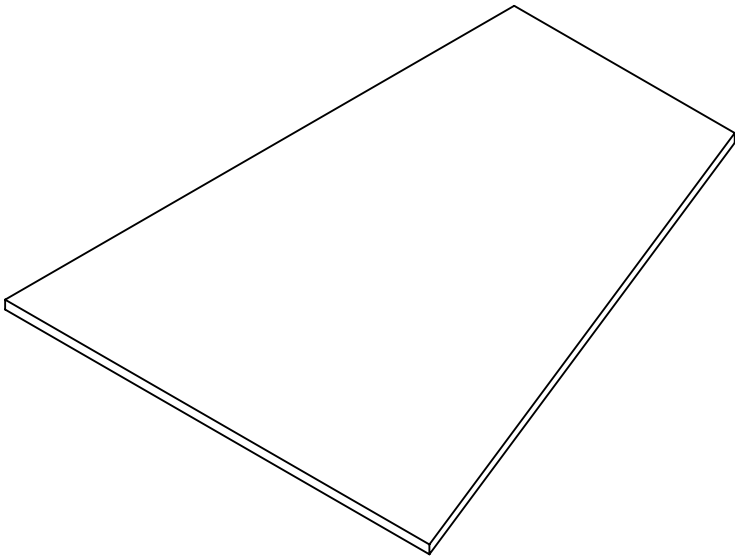
WIND TUNNEL / P14 / DIFFUSER BOTTOM

WT-P14 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

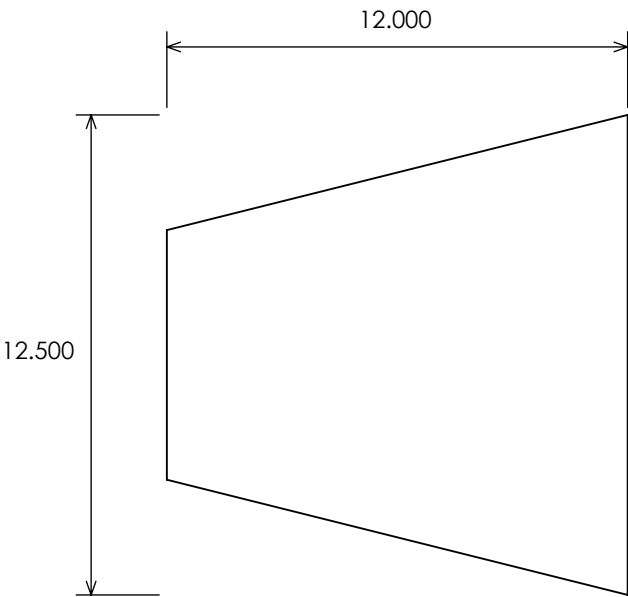
SIDE / X-Z | 1:3



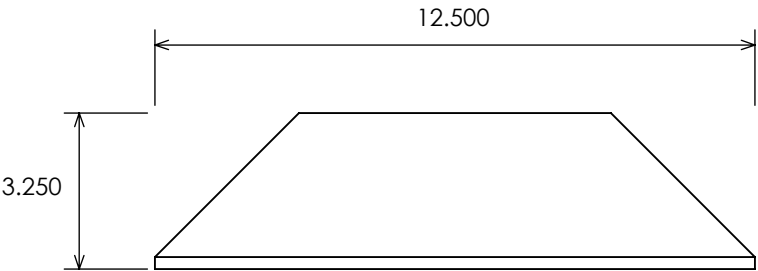
ISOMETRIC | 1:4



PLAN / X-Y | 1:5



END / Y-Z | 1:4



PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Entry panel width: 6.500
- Exit panel width: 12.500
- Radial wall thickness: 0.250
- Normal thickness: 0.242536
- Wall rise / fall: 3.000
- Wall slope: 14.0362 deg
- Centerline face length: 12.369317
- Inner edge seam: 12.727922
- Clear bore: 6.000 to 12.000 square
- Ends perpendicular to X axis.
- Lofted installed panel; not a flat pattern.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

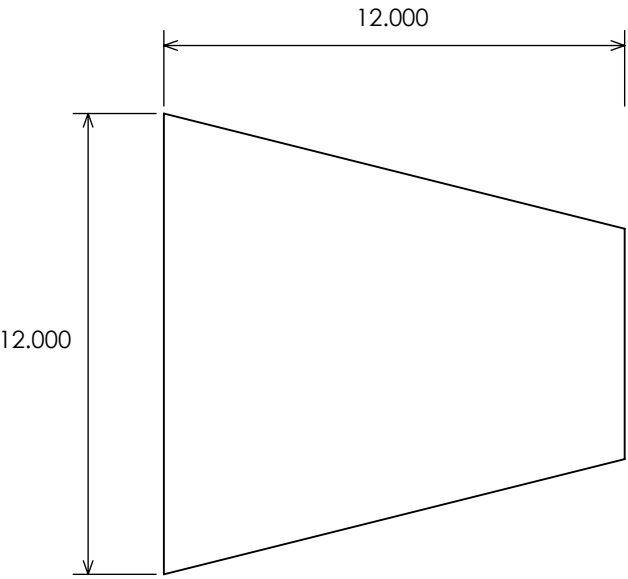
ASSEMBLY QTY: 1 | 04_Diffuser_Bottom.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

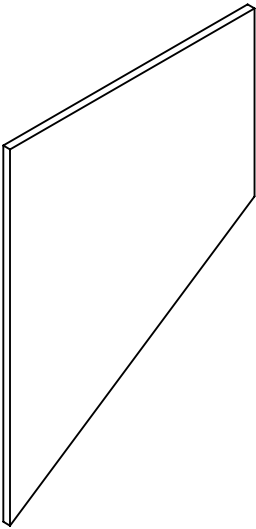
WIND TUNNEL / P15 / DIFFUSER RIGHT

WT-P15 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:5



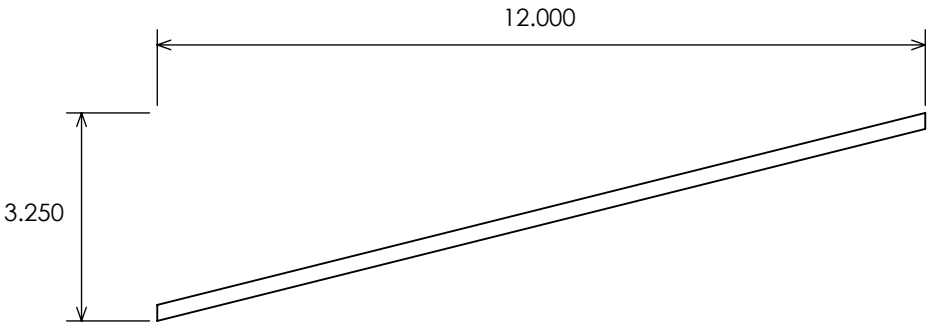
ISOMETRIC | 1:5



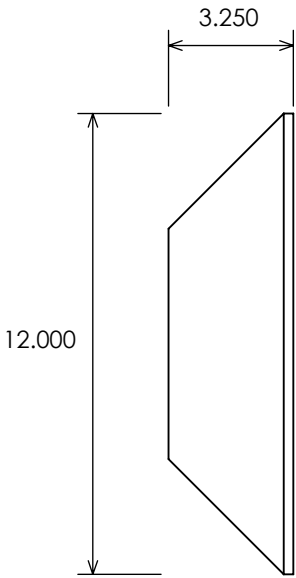
PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Entry panel width: 6.000
- Exit panel width: 12.000
- Radial wall thickness: 0.250
- Normal thickness: 0.242536
- Wall rise / fall: 3.000
- Wall slope: 14.0362 deg
- Centerline face length: 12.369317
- Inner edge seam: 12.727922
- Clear bore: 6.000 to 12.000 square
- Ends perpendicular to X axis.
- Lofted installed panel; not a flat pattern.

PLAN / X-Y | 1:3



END / Y-Z | 1:5



MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

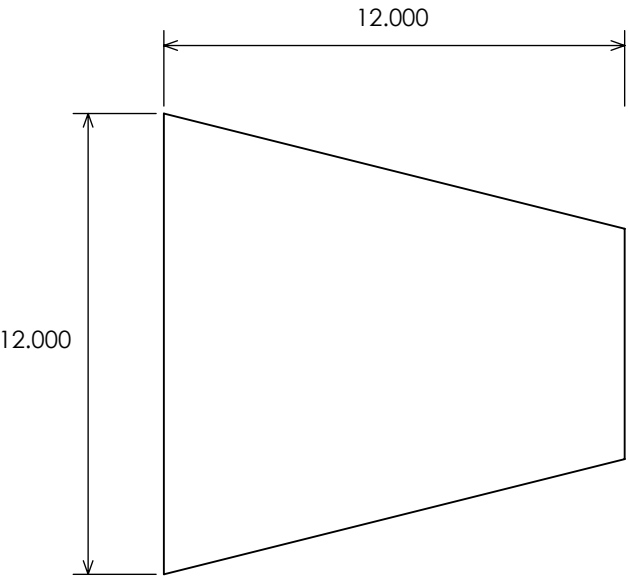
ASSEMBLY QTY: 1 | 04_Diffuser_Right.SLDprt

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

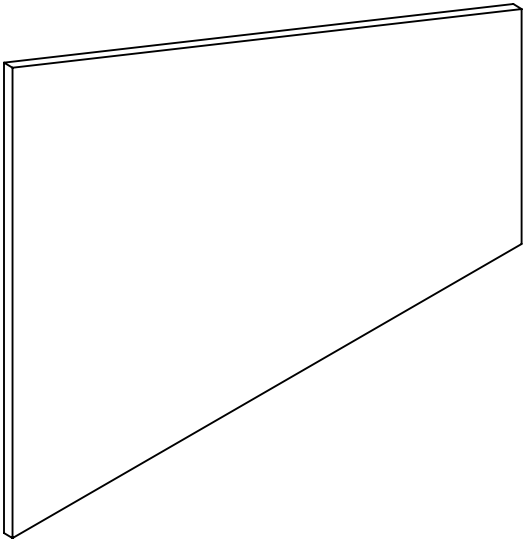
WIND TUNNEL / P16 / DIFFUSER LEFT

WT-P16 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:5



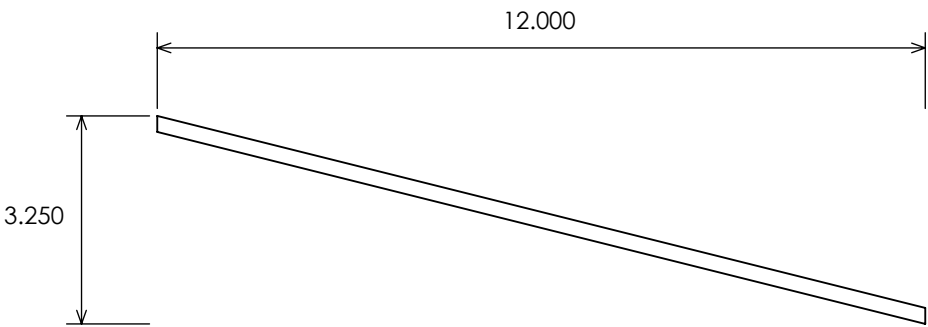
ISOMETRIC | 1:4



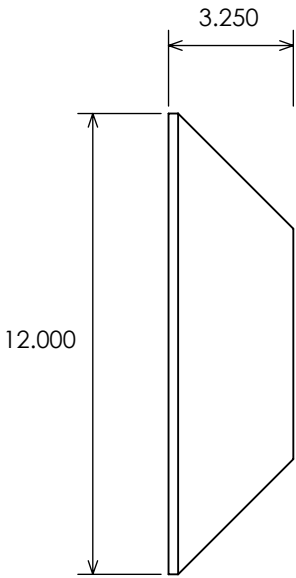
PART DIMENSIONS / DETAILS

- Axial length X: 12.000
- Entry panel width: 6.000
- Exit panel width: 12.000
- Radial wall thickness: 0.250
- Normal thickness: 0.242536
- Wall rise / fall: 3.000
- Wall slope: 14.0362 deg
- Centerline face length: 12.369317
- Inner edge seam: 12.727922
- Clear bore: 6.000 to 12.000 square
- Ends perpendicular to X axis.
- Lofted installed panel; not a flat pattern.

PLAN / X-Y | 1:3



END / Y-Z | 1:5



MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: BCX plywood - nominal

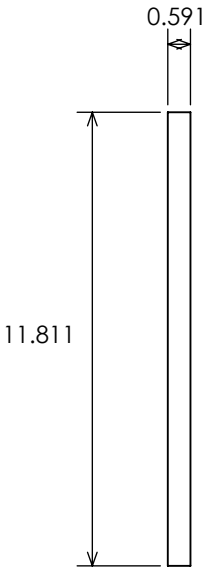
ASSEMBLY QTY: 1 | 04_Diffuser_Left.SLDPR

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

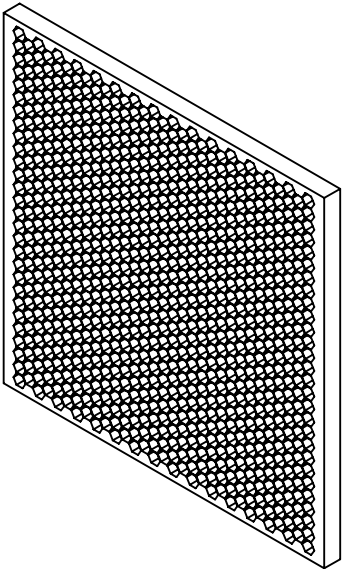
WIND TUNNEL / P17 / ARAMID HONEYCOMB

WT-P17 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:5



ISOMETRIC | 1:5



PART DIMENSIONS / DETAILS

ALL DETAIL VALUES BELOW IN mm

Outer square: 300 x 300

Axial thickness: 15

Hex opening vertex radius: 5.653590

Opening across flats: 9.792305

Opening across corners: 11.307180

Column pitch Y: 9

Row pitch Z: 10.392305

Odd columns stagger: 5.196152

Approx. shared wall: 0.600

Pore centers: $Y = 9c$

$Z = 10.392305(r + 0.5 \mid c \mid \bmod 2)$

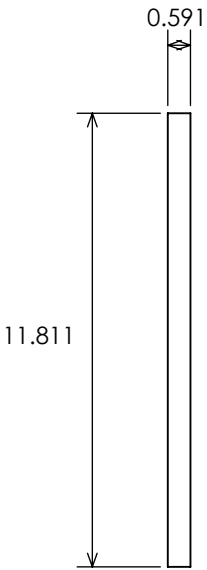
$c = -16..16; r = -15..15$

Vertices at 0,60,...300 degrees.

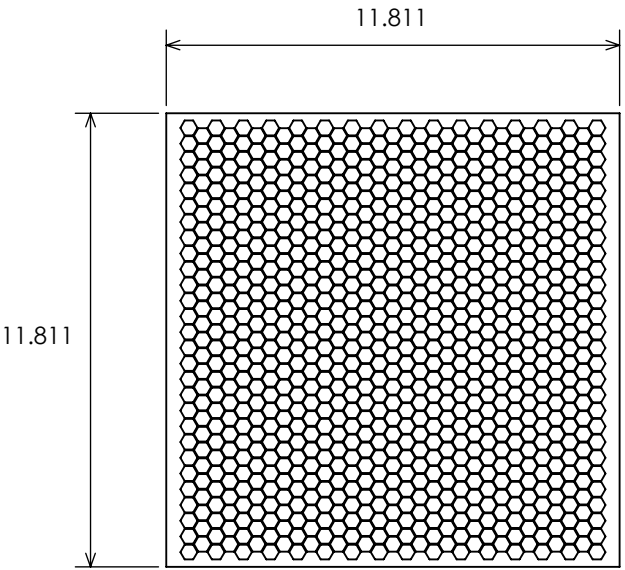
Keep pores with $\mid Y \mid, \mid Z \mid \leq 148$.

Pattern is representative, not measured.

PLAN / X-Y | 1:5



END / Y-Z | 1:5



MODEL AXES

X: extrusion / axial direction

Y: transverse; Z: vertical

Views independently scaled as labeled.

Overall dimensions rounded to 0.001 in.

Detail values control small features.

MATERIAL: Aramid paper - assumed

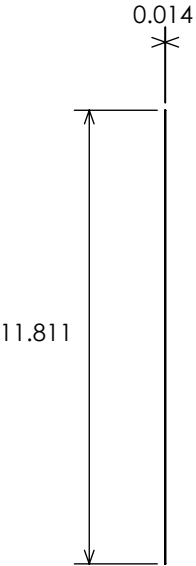
ASSEMBLY QTY: 1 | 05_Aramid_Honeycomb.SLDPR

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

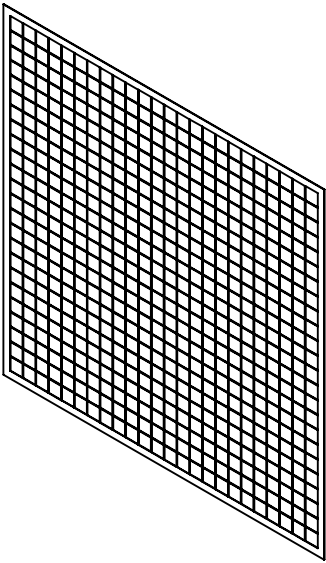
WIND TUNNEL / P18 / ALUMINUM SCREEN

WT-P18 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:5



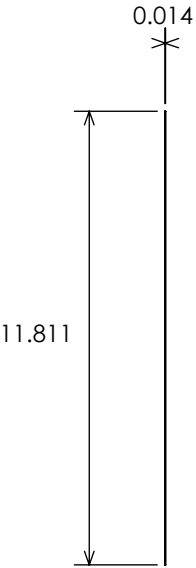
ISOMETRIC | 1:5



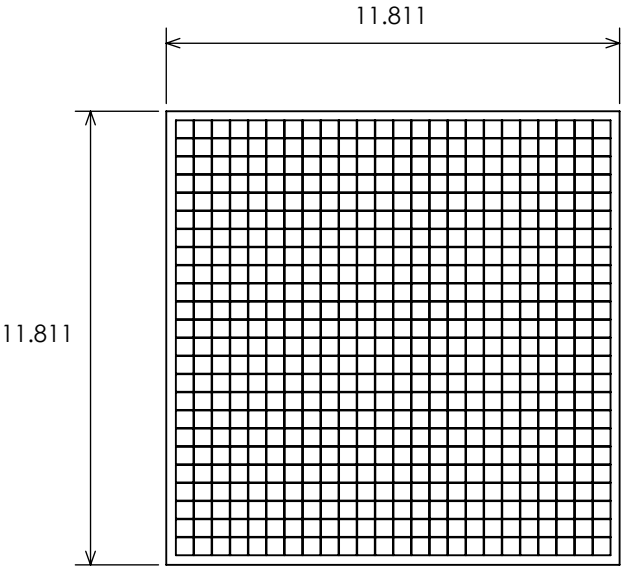
PART DIMENSIONS / DETAILS

ALL DETAIL VALUES BELOW IN mm
Outer square: 300 x 300
Thickness X: 0.350
576 square openings: 24 x 24
Each opening: 11.650 x 11.650
Pitch Y and Z: 12.000
Strip width: 0.350
Lower Y/Z border: 6.350
Upper Y/Z border: 6.000
Lower corner of first opening:
Y = -143.650; Z = -143.650
Other openings repeat at 12 mm.
Representative mesh; weave omitted.

PLAN / X-Y | 1:5



END / Y-Z | 1:5



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: 1060 Alloy

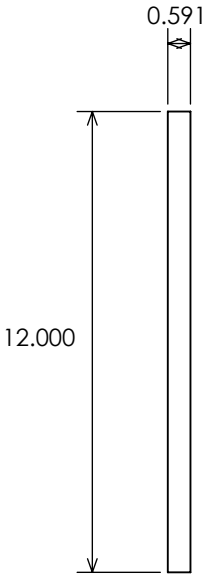
ASSEMBLY QTY: 2 | 05_Aluminum_Screen.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

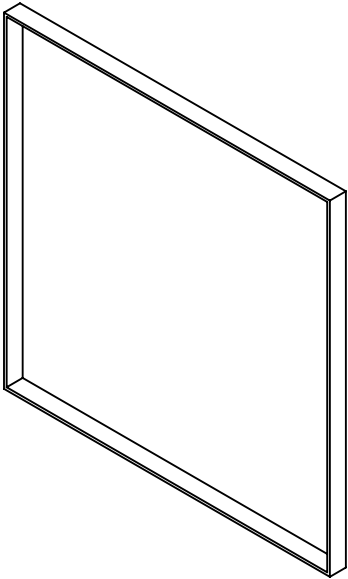
WIND TUNNEL / P19 / HONEYCOMB EDGE SEAL

WT-P19 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

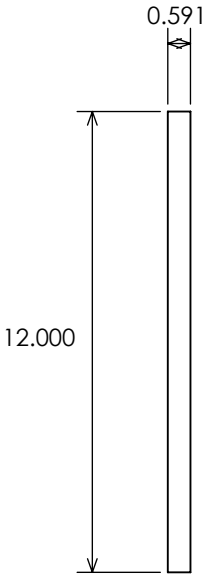
SIDE / X-Z | 1:5



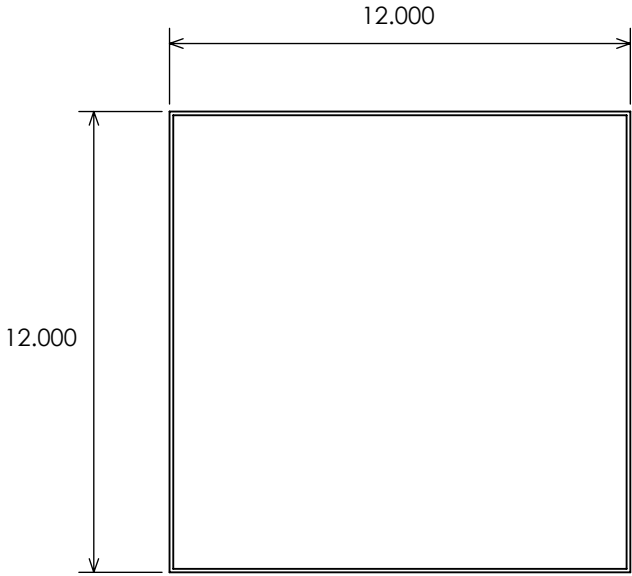
ISOMETRIC | 1:5



PLAN / X-Y | 1:5



END / Y-Z | 1:5



PART DIMENSIONS / DETAILS

- Outer square: 12.000
- Inner square: 300 mm (11.811024)
- Axial thickness: 15 mm (0.590551)
- Seal wall: 2.4 mm (0.094488)
- Open center; continuous perimeter.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: Foam weatherstrip - assumed

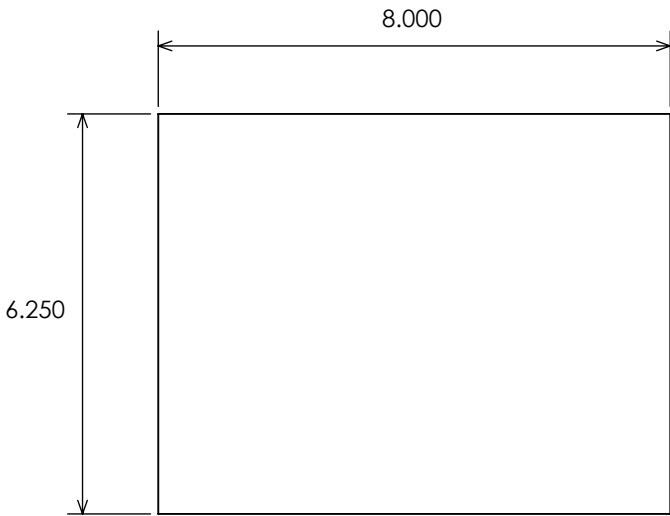
ASSEMBLY QTY: 1 | 05_Honeycomb_Edge_Seal.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

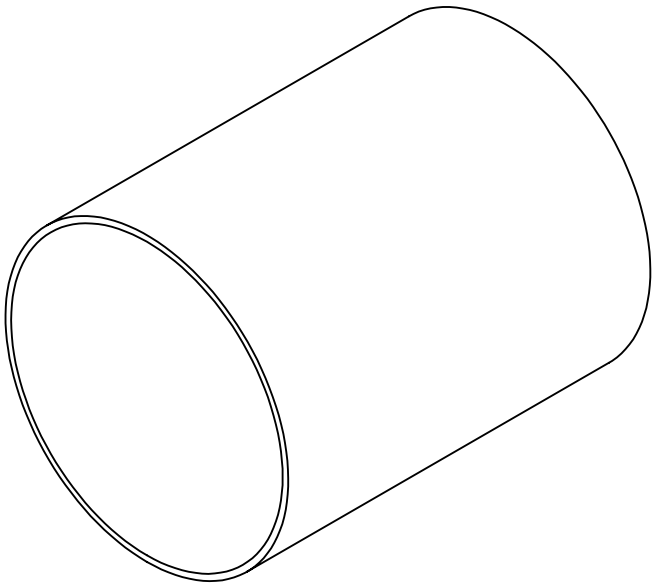
WIND TUNNEL / P20 / FAN HOUSING

WT-P20 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:3



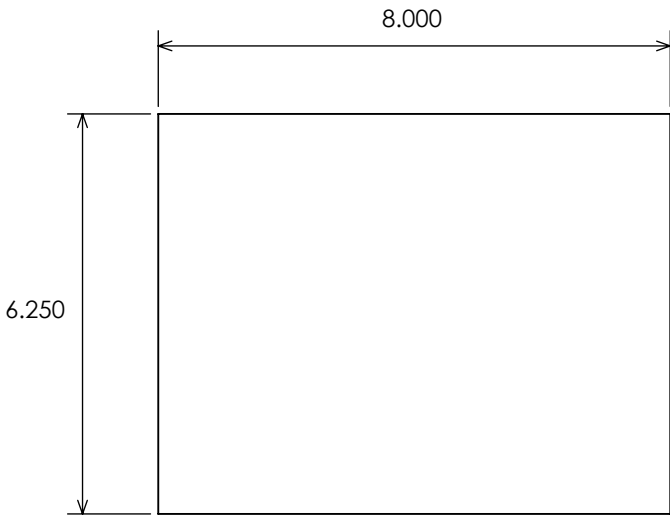
ISOMETRIC | 1:3



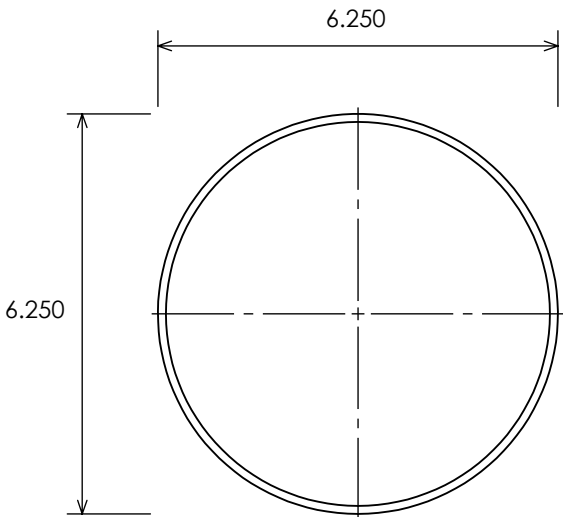
PART DIMENSIONS / DETAILS

- Axial length: 8.000
- Outer diameter: 6.250
- Inner diameter: 6.000
- Wall thickness: 0.125
- Open at both ends.
- Purchased fan envelope only.

PLAN / X-Y | 1:3



END / Y-Z | 1:3



MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: Plain Carbon Steel

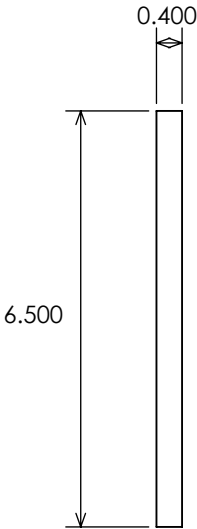
ASSEMBLY QTY: 1 | 06_Fan_Housing.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

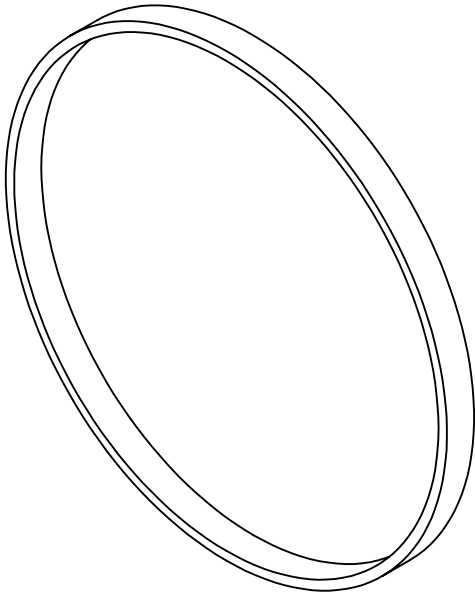
WIND TUNNEL / P21 / FAN END COLLAR

WT-P21 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:3



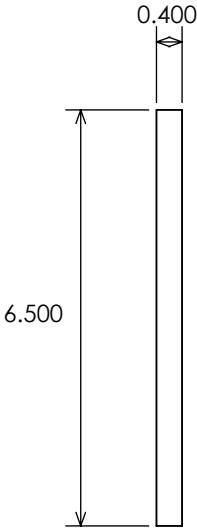
ISOMETRIC | 1:2



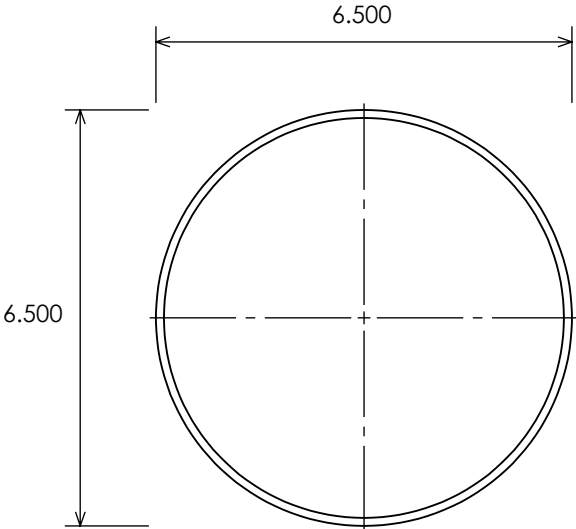
PART DIMENSIONS / DETAILS

Axial length: 0.400
Outer diameter: 6.500
Inner diameter: 6.250
Radial wall: 0.125

PLAN / X-Y | 1:3



END / Y-Z | 1:3



MODEL AXES

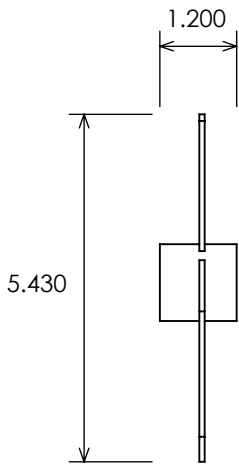
X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Plain Carbon Steel

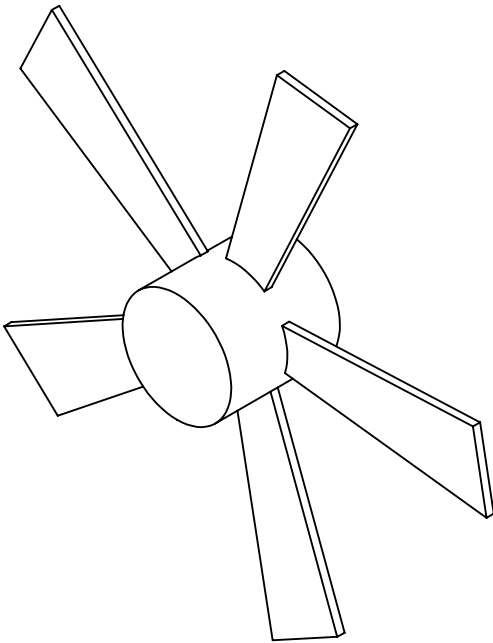
ASSEMBLY QTY: 2 | 06_Fan_End_Collar.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

SIDE / X-Z | 1:3



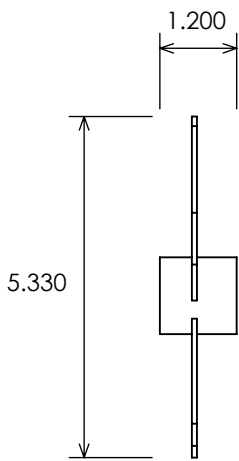
ISOMETRIC | 1:1.5



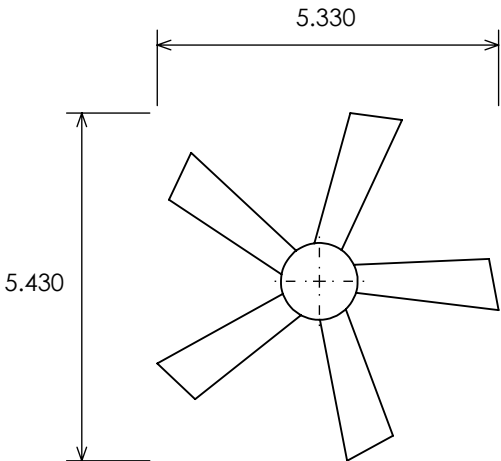
PART DIMENSIONS / DETAILS

Hub diameter: 1.200
Hub axial length: 1.200
Five blades at 72 deg spacing
Blade thickness X: 0.080
Blade plane: X = 0.500 to 0.580
First blade outline (Y, Z):
A = (0.350, -0.150)
B = (2.800, -0.450)
C = (2.650, 0.350)
D = (0.350, 0.250)
Rotate outline about X for copies.
Flat static blades; pitch omitted.
Entire impeller geometry assumed.

PLAN / X-Y | 1:3



END / Y-Z | 1:3



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

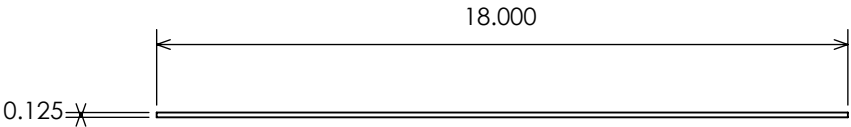
MATERIAL: ABS
ASSEMBLY QTY: 1 | 06_Fan_Impeller.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

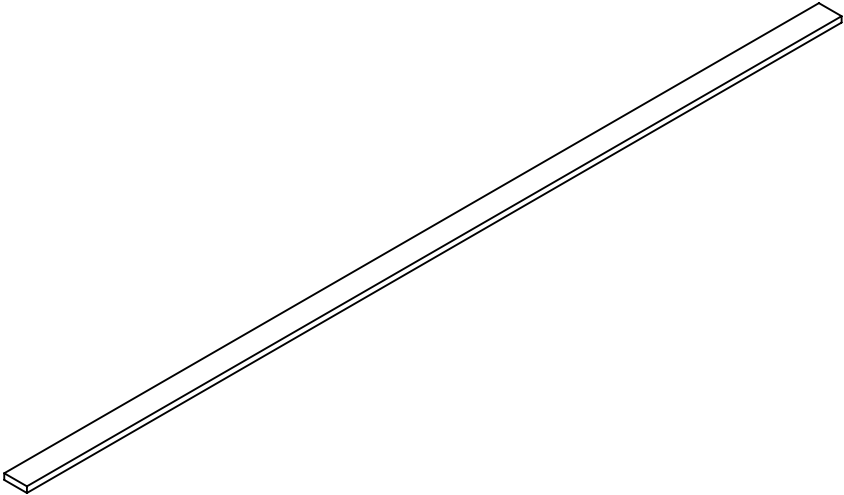
WIND TUNNEL / P23 / LID GASKET RAIL

WT-P23 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:5



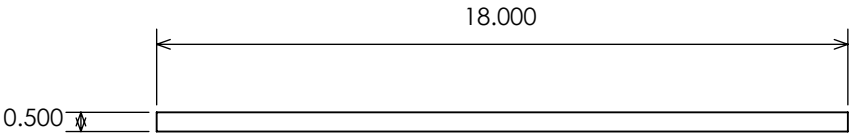
ISOMETRIC | 1:3



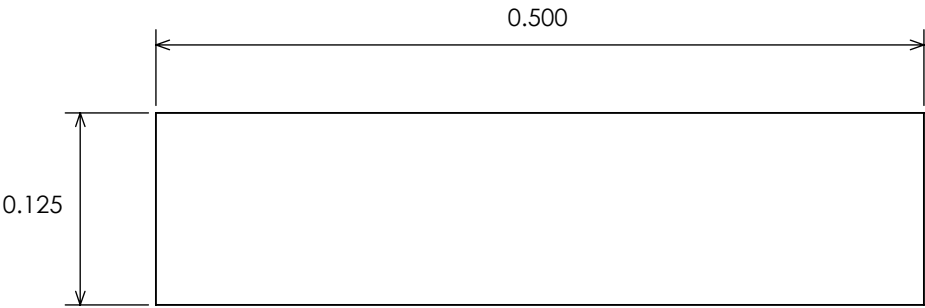
PART DIMENSIONS / DETAILS

Length X: 18.000
Width Y: 0.500
Height Z: 0.125
Nominal uncompressed weatherstrip.

PLAN / X-Y | 1:5



END / Y-Z | 8:1



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Foam weatherstrip - assumed

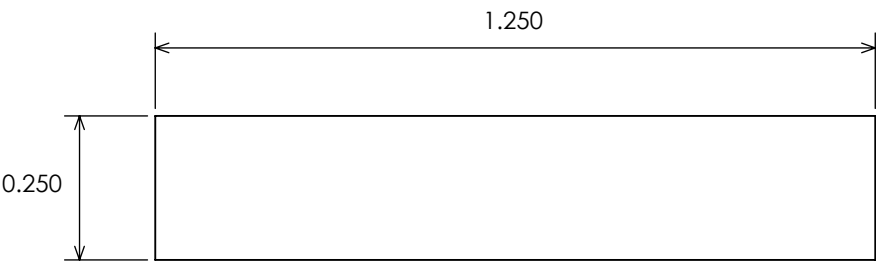
ASSEMBLY QTY: 2 | 07_Lid_Gasket_Rail.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

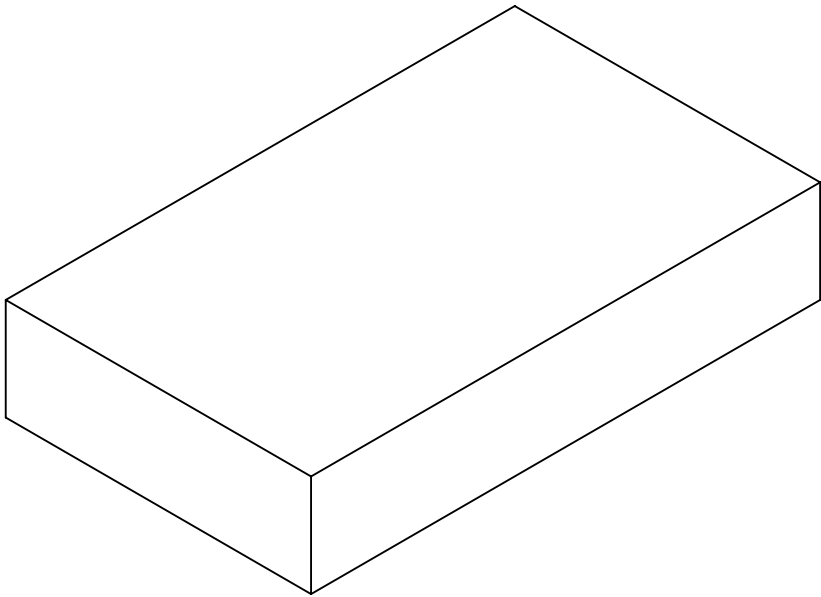
WIND TUNNEL / P24 / MAGNET SUPPORT TAB

WT-P24 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

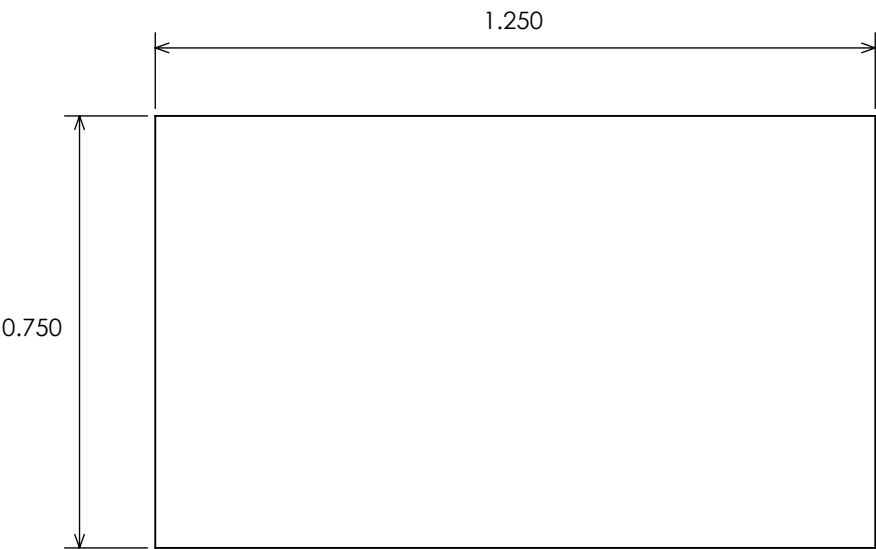
SIDE / X-Z | 3:1



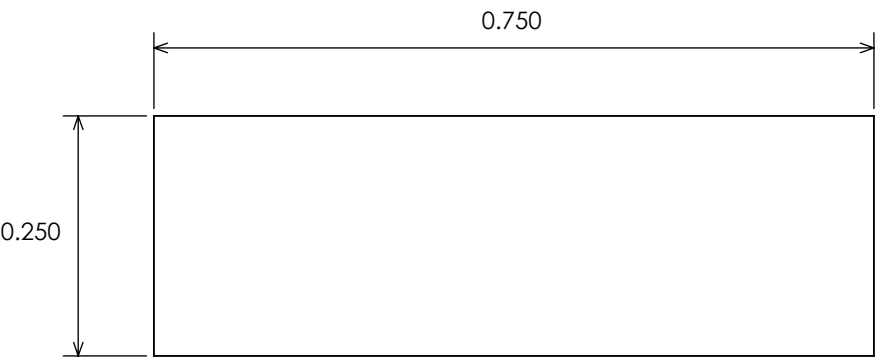
ISOMETRIC | 3:1



PLAN / X-Y | 3:1



END / Y-Z | 5:1



PART DIMENSIONS / DETAILS

Length X: 1.250
Width Y: 0.750
Thickness Z: 0.250
Attachment holes not modeled.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: BCX plywood - nominal

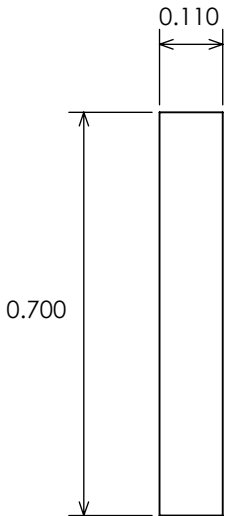
ASSEMBLY QTY: 4 | 07_Magnet_Support_Tab.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

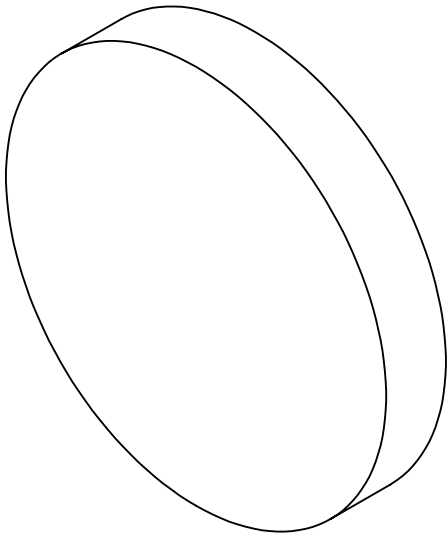
WIND TUNNEL / P25 / DISC MAGNET

WT-P25 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

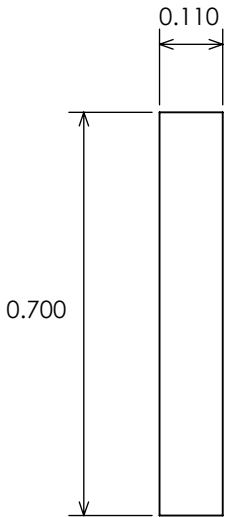
SIDE / X-Z | 3:1



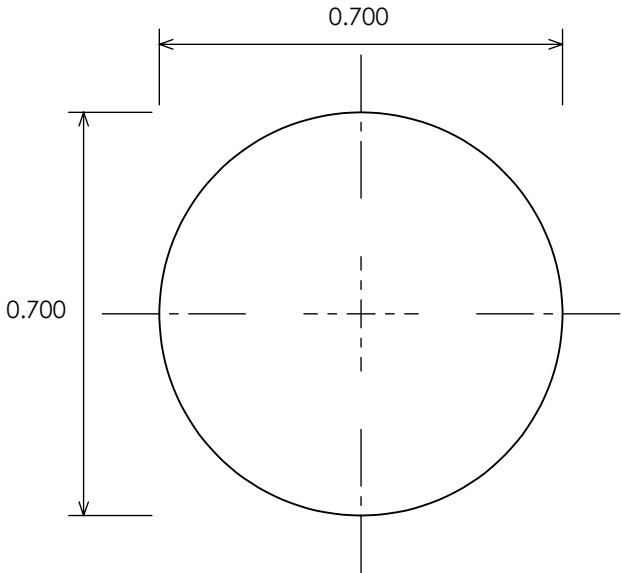
ISOMETRIC | 4:1



PLAN / X-Y | 3:1



END / Y-Z | 3:1



PART DIMENSIONS / DETAILS

Diameter: 0.700
Thickness X: 0.110
Eight magnets / four pairs.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Neodymium magnet - assumed

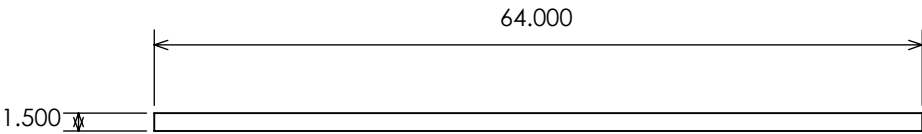
ASSEMBLY QTY: 8 | 07_Disc_Magnet.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

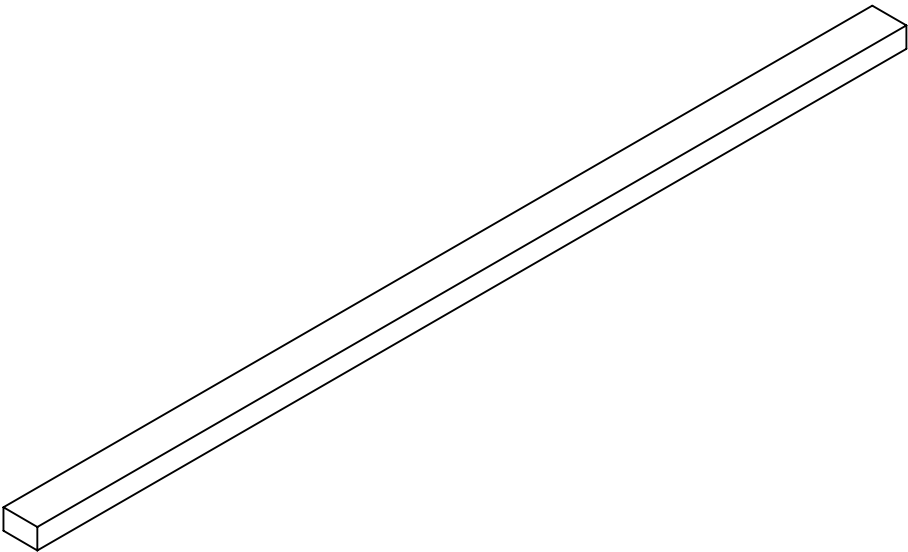
WIND TUNNEL / P26 / BASE RAIL

WT-P26 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:16



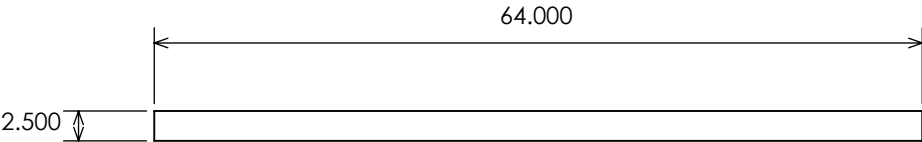
ISOMETRIC | 1:10



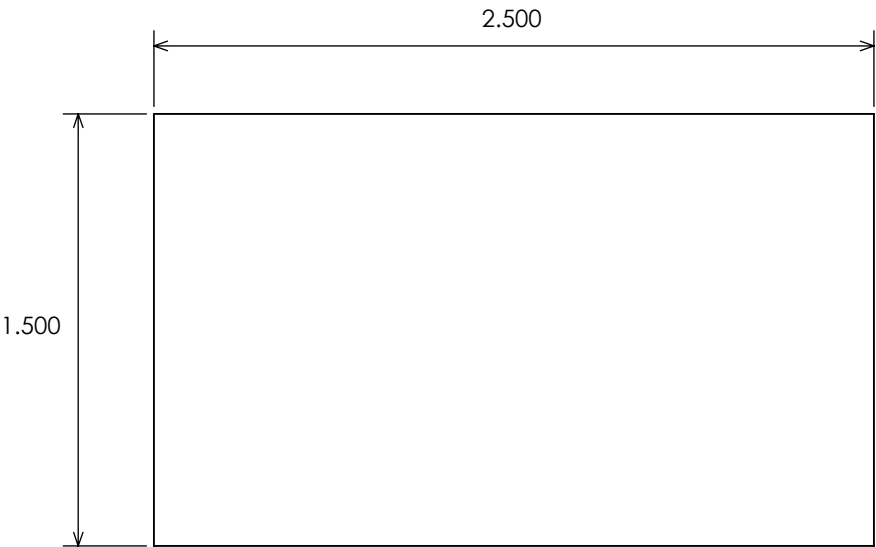
PART DIMENSIONS / DETAILS

Length X: 64.000
Width Y: 2.500
Height Z: 1.500
Dressed nominal 2 x 3 lumber.

PLAN / X-Y | 1:16



END / Y-Z | 1.5:1



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Pine

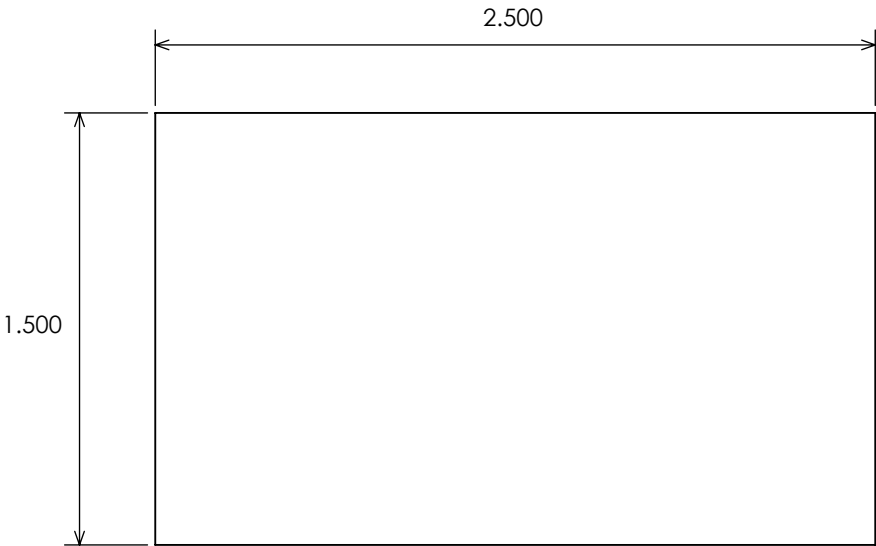
ASSEMBLY QTY: 2 | 08_Base_Rail.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

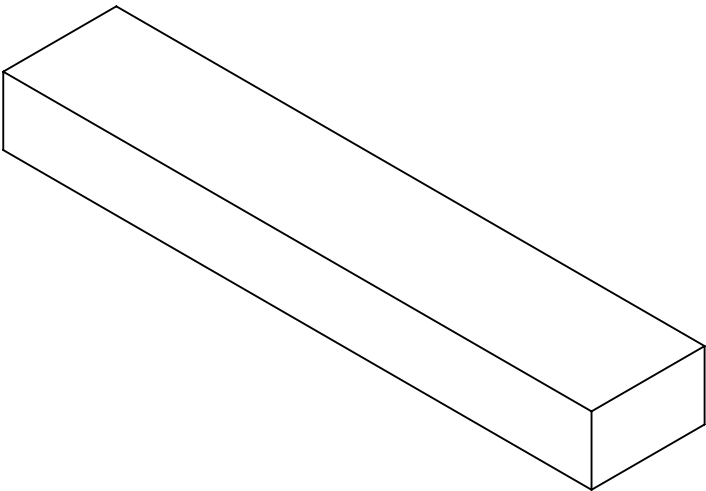
WIND TUNNEL / P27 / CROSSBEAM

WT-P27 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1.5:1



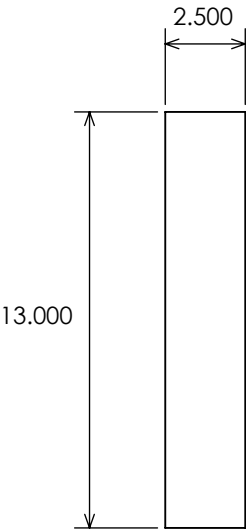
ISOMETRIC | 1:3



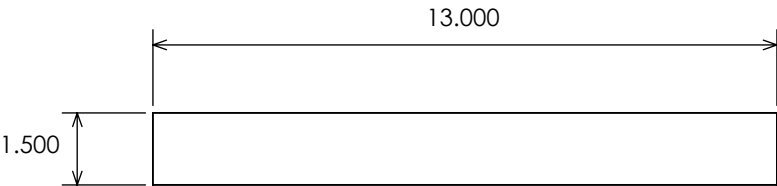
PART DIMENSIONS / DETAILS

Depth X: 2.500
Span Y: 13.000
Height Z: 1.500
Dressed nominal 2 x 3 lumber.

PLAN / X-Y | 1:6



END / Y-Z | 1:4



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Pine

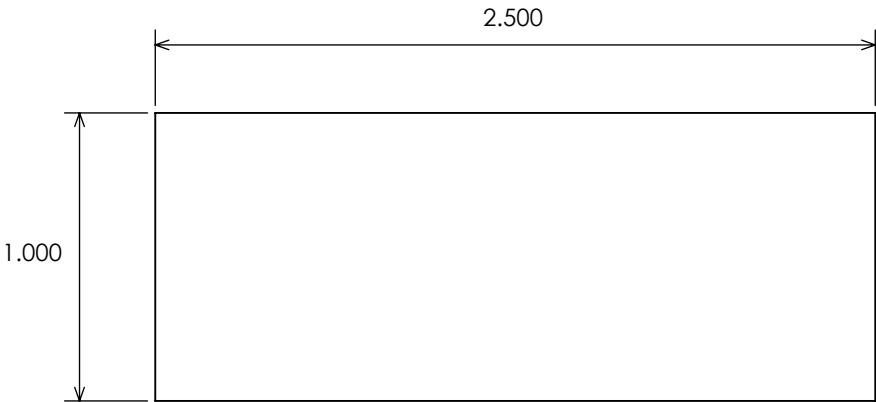
ASSEMBLY QTY: 5 | 08_Crossbeam.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

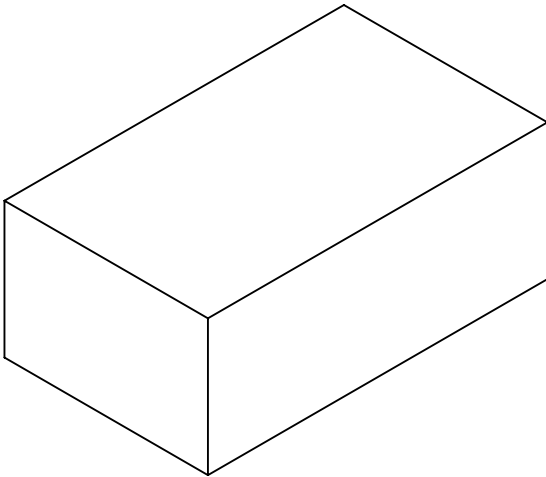
WIND TUNNEL / P28 / INLET DIFFUSER POST

WT-P28 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1.5:1



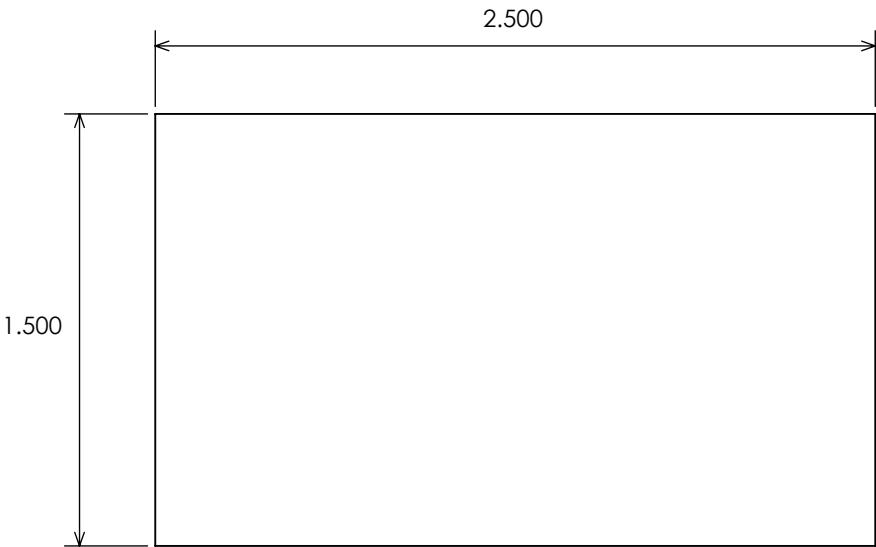
ISOMETRIC | 1:1



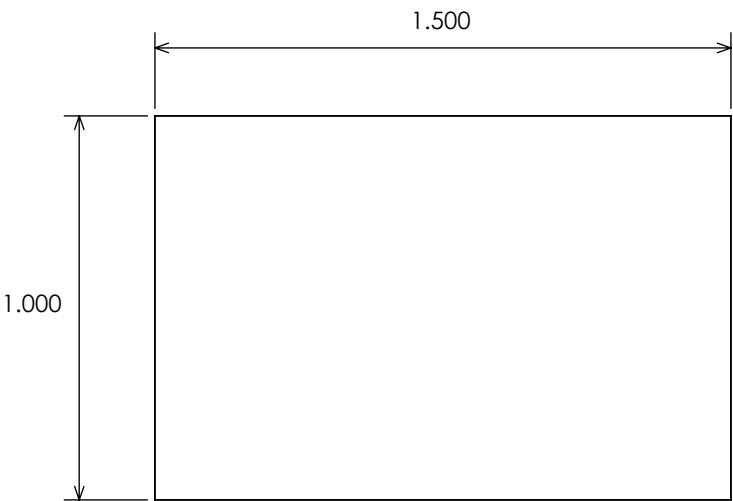
PART DIMENSIONS / DETAILS

Depth X: 2.500
Width Y: 1.500
Height Z: 1.000
Assumed support geometry.

PLAN / X-Y | 1.5:1



END / Y-Z | 2:1



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Pine

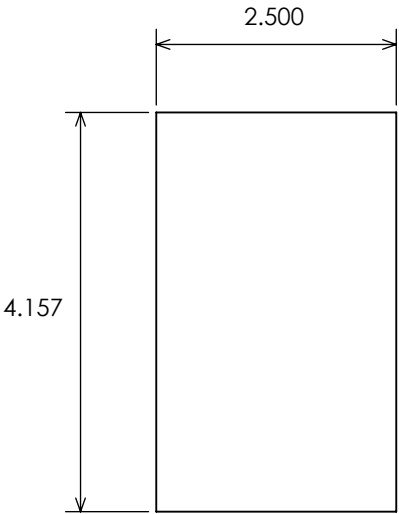
ASSEMBLY QTY: 4 | 08_Inlet_Diffuser_Post.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

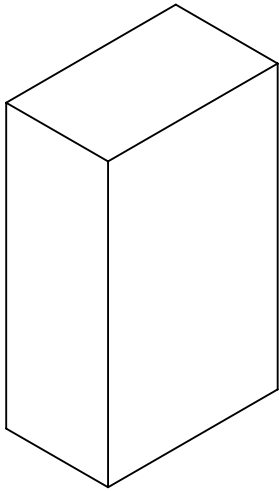
WIND TUNNEL / P29 / TEST SUPPORT POST

WT-P29 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:2



ISOMETRIC | 1:2



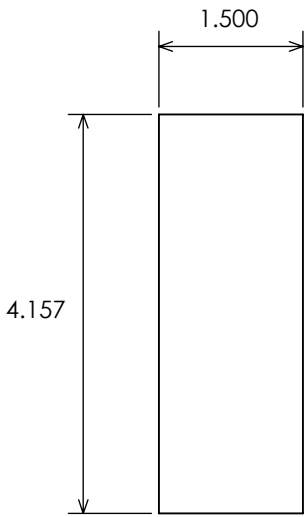
PART DIMENSIONS / DETAILS

Depth X: 2.500
Width Y: 1.500
Height Z: 4.157
Assumed support geometry.

PLAN / X-Y | 1.5:1



END / Y-Z | 1:2



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

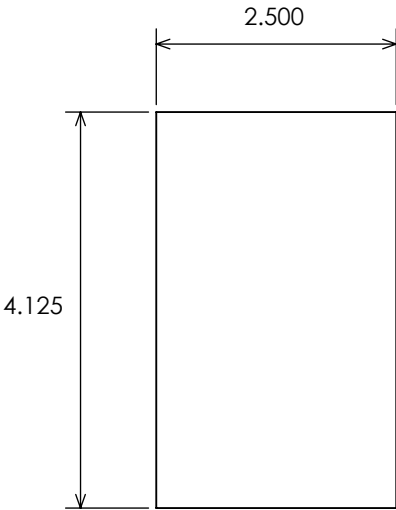
MATERIAL: Pine
ASSEMBLY QTY: 4 | 08_Test_Support_Post.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

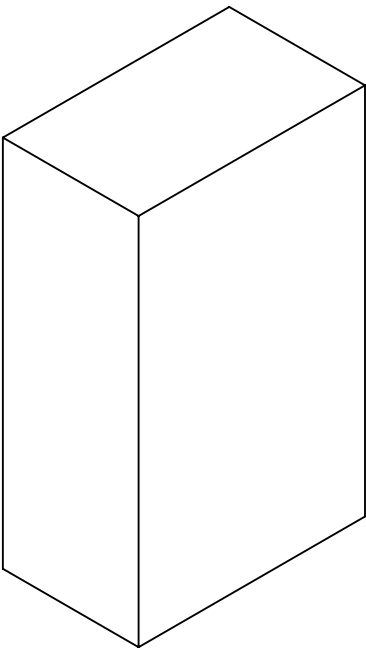
WIND TUNNEL / P30 / FAN SUPPORT POST

WT-P30 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:2



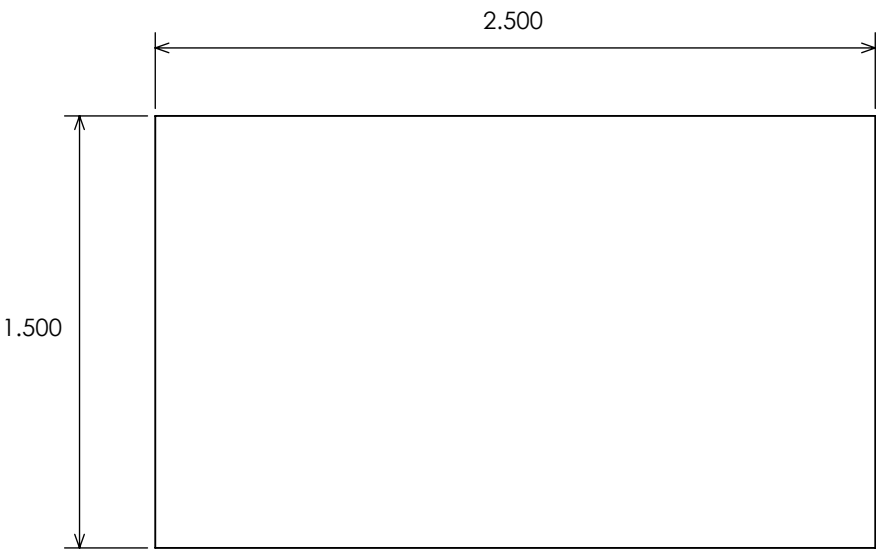
ISOMETRIC | 1:1.5



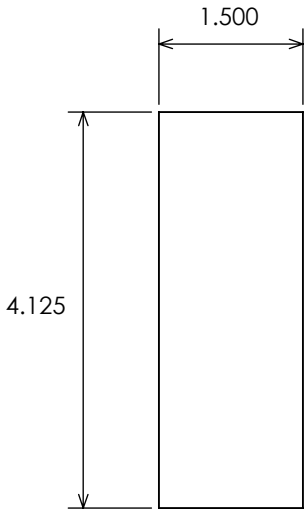
PART DIMENSIONS / DETAILS

Depth X: 2.500
Width Y: 1.500
Height Z: 4.125
Assumed support geometry.

PLAN / X-Y | 1.5:1



END / Y-Z | 1:2



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Pine

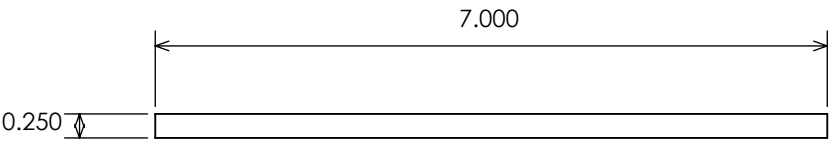
ASSEMBLY QTY: 2 | 08_Fan_Support_Post.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

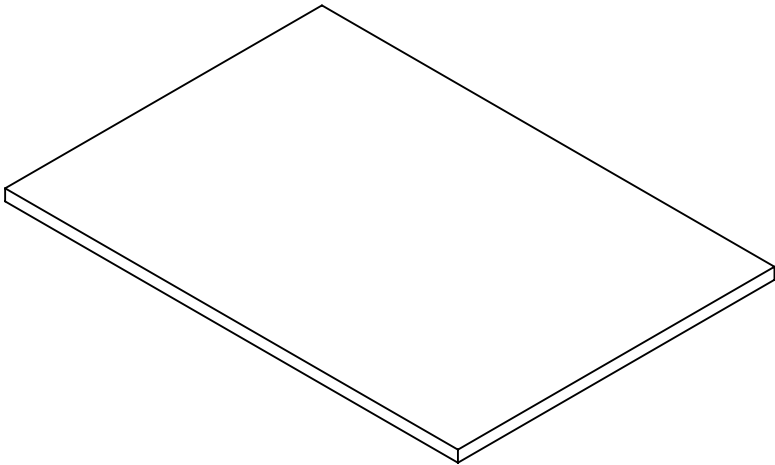
WIND TUNNEL / P31 / ELECTRONICS SHELF

WT-P31 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:2



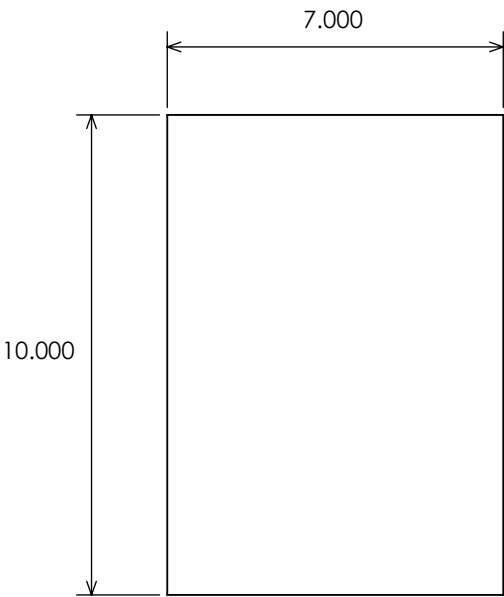
ISOMETRIC | 1:3



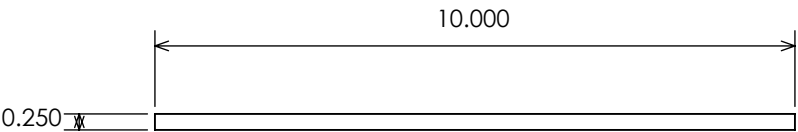
PART DIMENSIONS / DETAILS

Length X: 7.000
Width Y: 10.000
Thickness Z: 0.250

PLAN / X-Y | 1:4



END / Y-Z | 1:3



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: BCX plywood - nominal

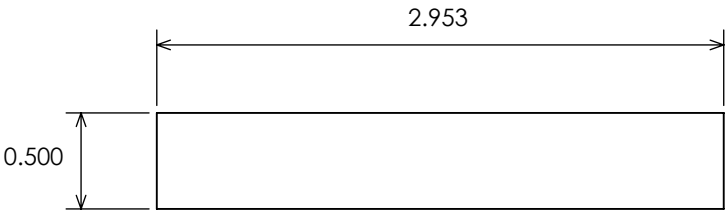
ASSEMBLY QTY: 1 | 08_Electronics_Shelf.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

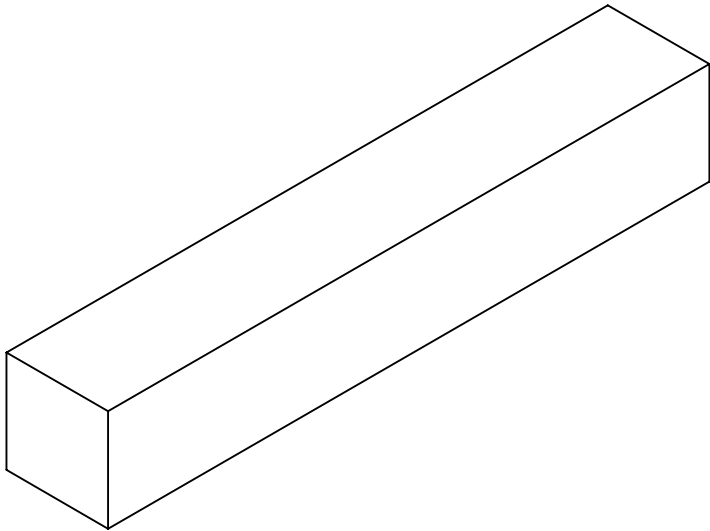
WIND TUNNEL / P32 / LOAD CELL 5KG

WT-P32 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

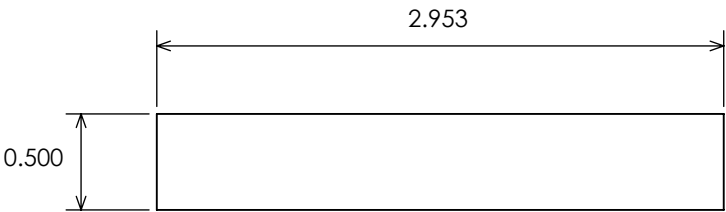
SIDE / X-Z | 1:1



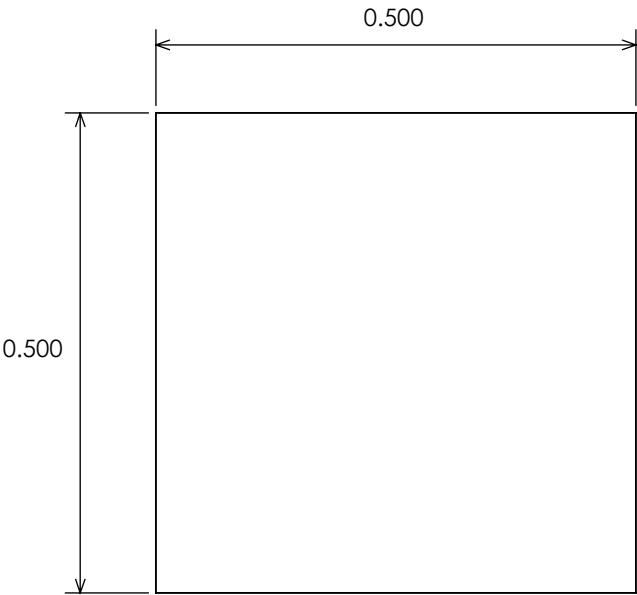
ISOMETRIC | 1.5:1



PLAN / X-Y | 1:1



END / Y-Z | 5:1



PART DIMENSIONS / DETAILS

ALL DETAIL VALUES BELOW IN mm

Length X: 75.000

Width Y: 12.700

Height Z: 12.700

5 kg purchased load-cell envelope.

Mounting holes / flexures omitted.

MODEL AXES

X: extrusion / axial direction

Y: transverse; Z: vertical

Views independently scaled as labeled.

Overall dimensions rounded to 0.001 in.

Detail values control small features.

MATERIAL: 1060 Alloy

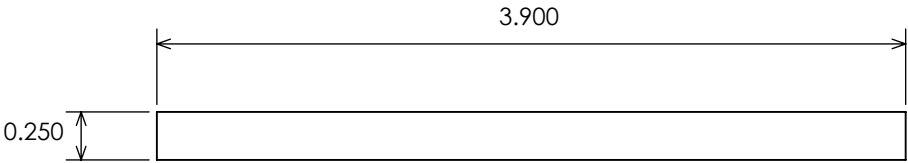
ASSEMBLY QTY: 1 | 09_Load_Cell_5kg.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

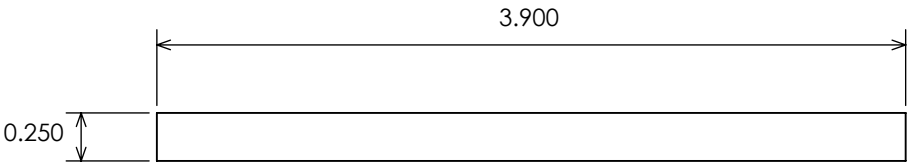
WIND TUNNEL / P33 / STAINLESS MODEL ROD

WT-P33 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

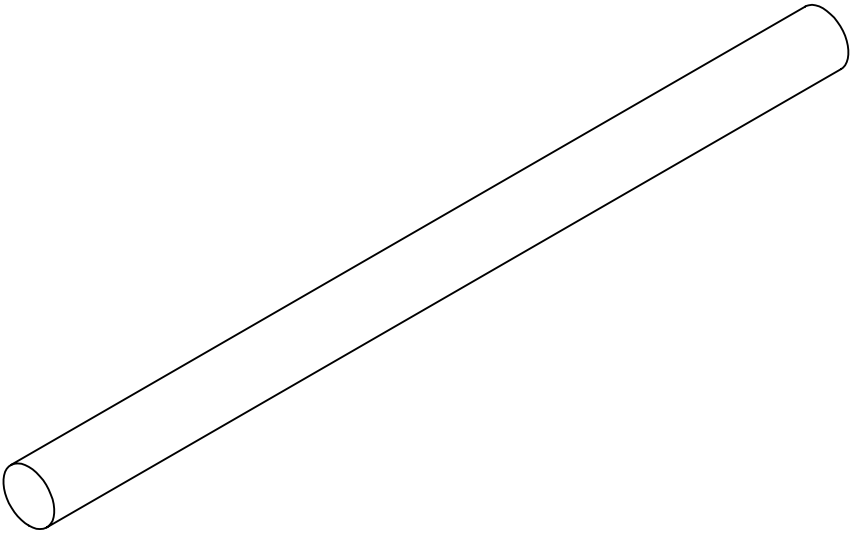
SIDE / X-Z | 1:1



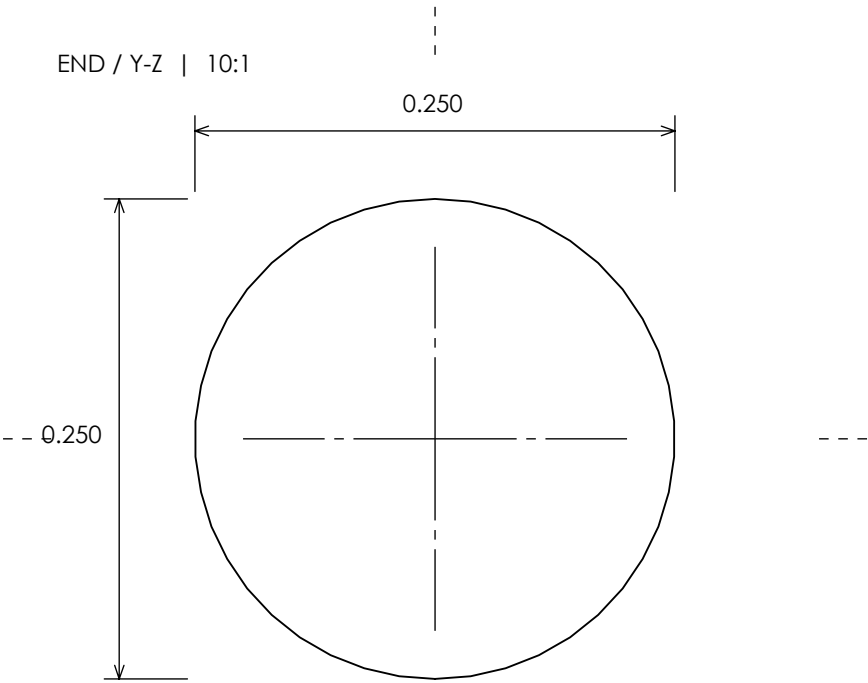
PLAN / X-Y | 1:1



ISOMETRIC | 1.5:1



END / Y-Z | 10:1



PART DIMENSIONS / DETAILS

Diameter: 0.250
Cut length X: 3.900
Rod stock diameter from BOM.
Cut length assumed.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: AISI 304

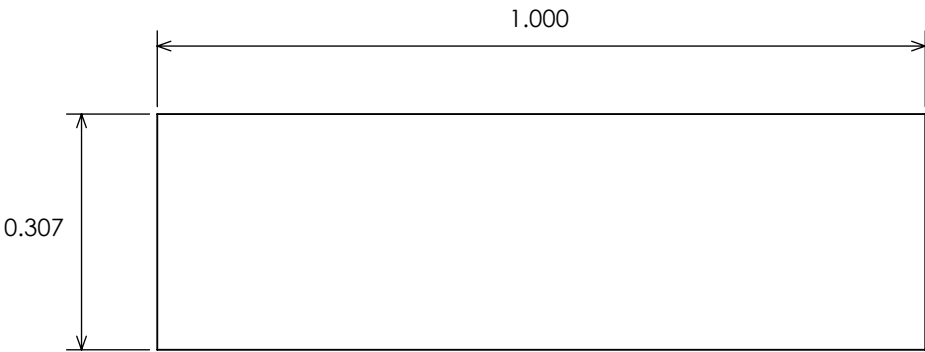
ASSEMBLY QTY: 1 | 09_Stainless_Model_Rod.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

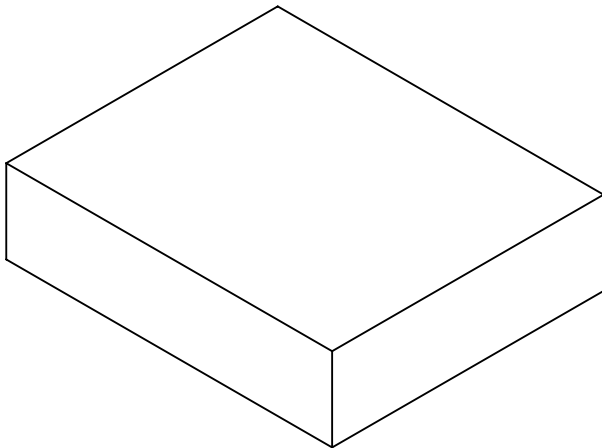
WIND TUNNEL / P34 / LOAD CELL MOUNT

WT-P34 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

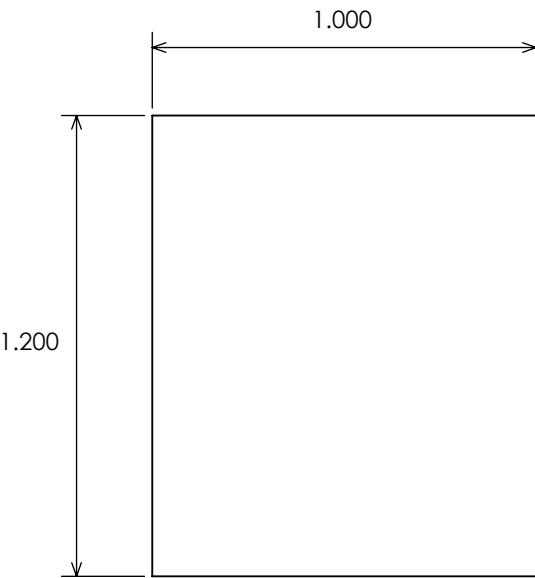
SIDE / X-Z | 4:1



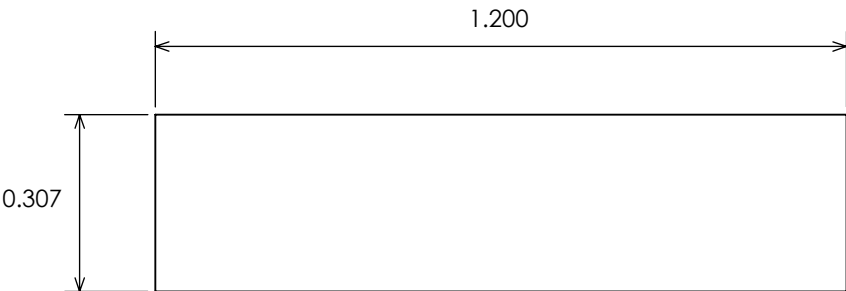
ISOMETRIC | 2:1



PLAN / X-Y | 2:1



END / Y-Z | 3:1



PART DIMENSIONS / DETAILS

Length X: 1.000
Width Y: 1.200
Height Z: 0.307
Attachment details not measured.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: BCX plywood - nominal

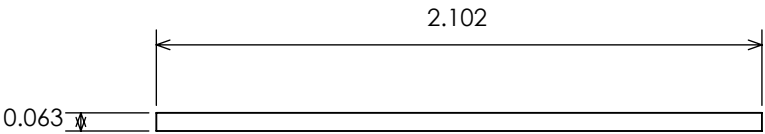
ASSEMBLY QTY: 1 | 09_Load_Cell_Mount.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

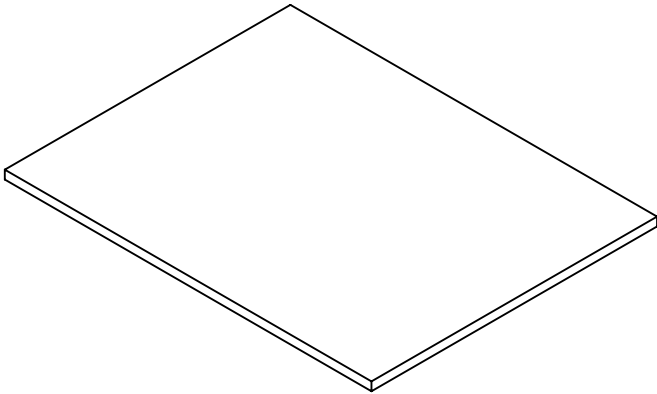
WIND TUNNEL / P35 / ARDUINO UNO BOARD

WT-P35 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1.5:1



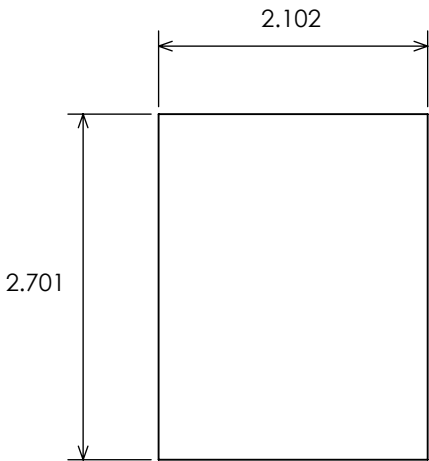
ISOMETRIC | 1:1



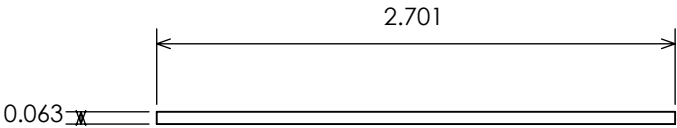
PART DIMENSIONS / DETAILS

ALL DETAIL VALUES BELOW IN mm
Length X: 53.400
Width Y: 68.600
Thickness Z: 1.600
UNO R3 envelope only.
Connectors / mounting holes omitted.

PLAN / X-Y | 1:1.5



END / Y-Z | 1:1



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: PCB laminate - assumed

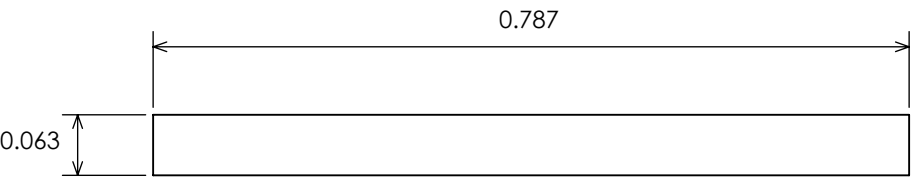
ASSEMBLY QTY: 1 | 09_Arduino_UNO_Board.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

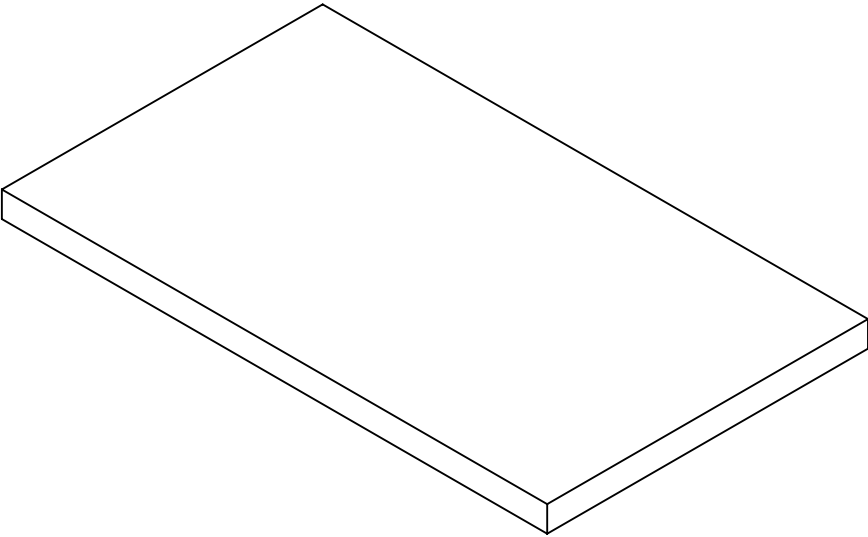
WIND TUNNEL / P36 / HX711 BOARD

WT-P36 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

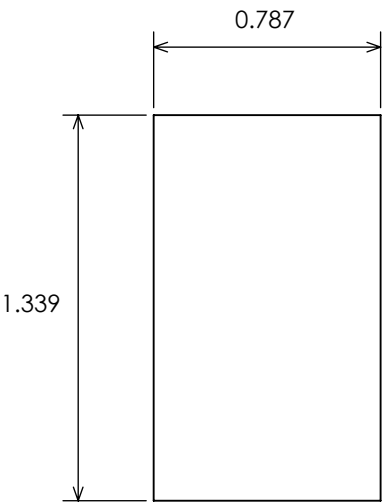
SIDE / X-Z | 5:1



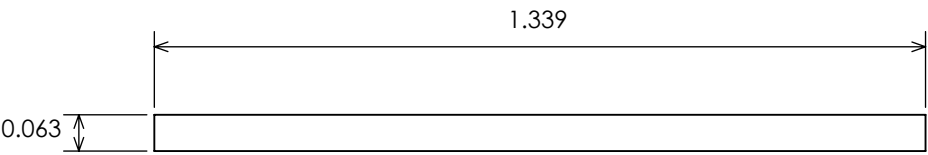
ISOMETRIC | 3:1



PLAN / X-Y | 1.5:1



END / Y-Z | 3:1



PART DIMENSIONS / DETAILS

ALL DETAIL VALUES BELOW IN mm
Length X: 20.000
Width Y: 34.000
Thickness Z: 1.600
HX711 envelope only.
Electrical features omitted.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: PCB laminate - assumed

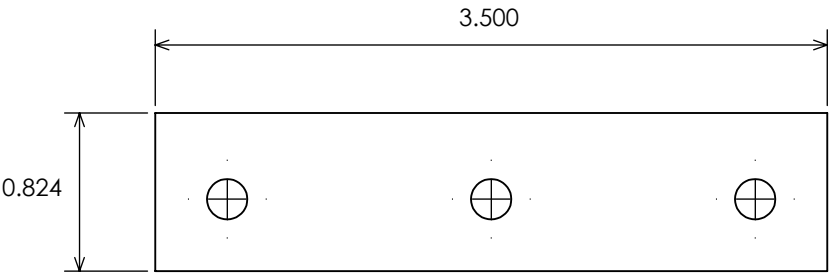
ASSEMBLY QTY: 1 | 09_HX711_Board.SLDPR

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

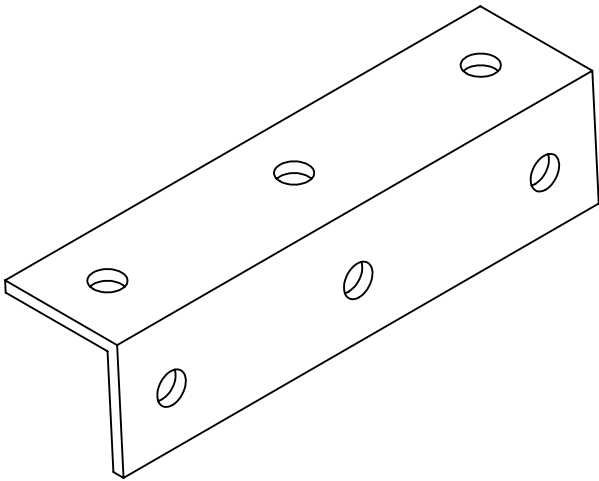
WIND TUNNEL / P37 / CORNER BRACE 3 5IN

WT-P37 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

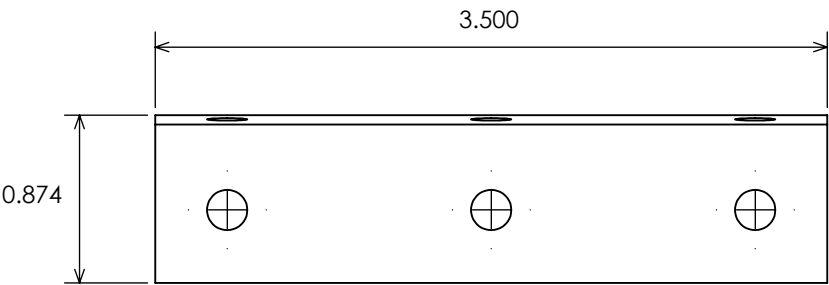
SIDE / X-Z | 1:1



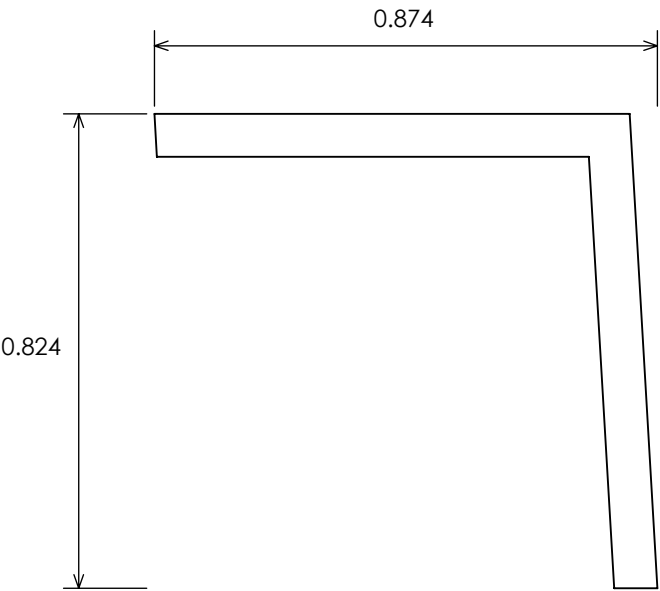
ISOMETRIC | 1:1



PLAN / X-Y | 1:1



END / Y-Z | 3:1



PART DIMENSIONS / DETAILS

Seam length X: 3.500
Flange leg length: 0.750 each
Nominal section gauge: 0.075
Inside bend angle: 93.372287 deg
6 x DIA 0.210 THRU
X hole stations from end:
0.375 / 1.750 / 3.125
Top flange hole center: Y = -0.375
Side flange: Z = -0.374351
Section profile in (u,v):
(-.750,0), (0,0), (0,-.750),
(.075,-.750), (.075,.075),
(-.750,.075). Sharp bend modeled.
 $Y = u - v/17$; $Z = 0.998268v$
Hole axes follow CAD Y and Z.
Opened bend is a fit assumption.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Galvanized Steel

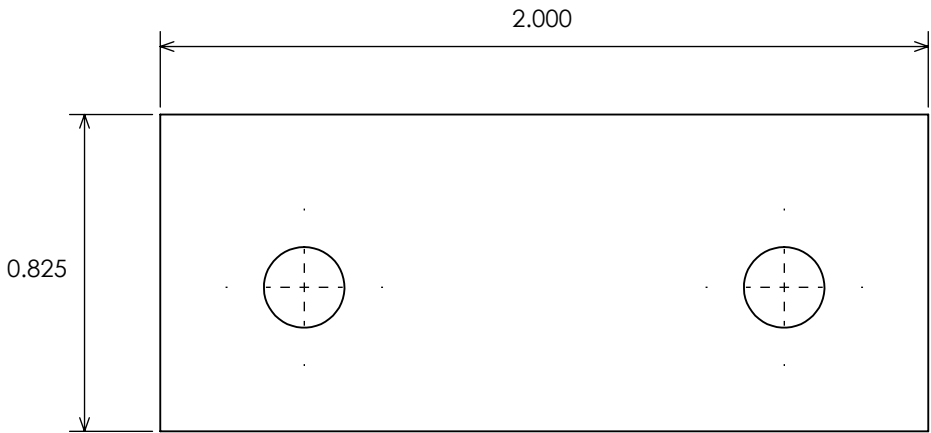
ASSEMBLY QTY: 16 | 10_Corner_Brace_3_5in.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

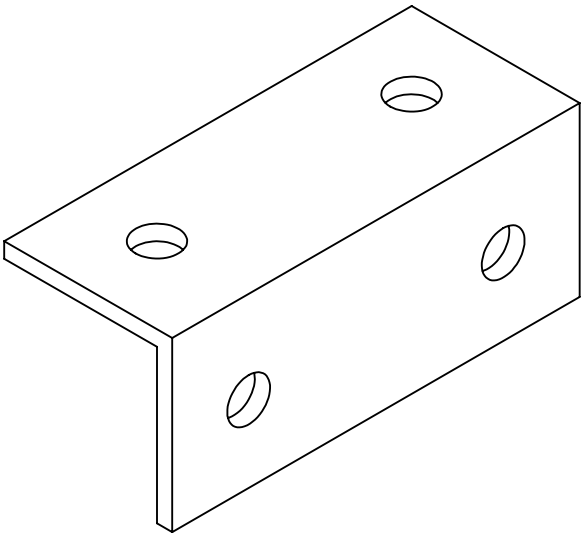
WIND TUNNEL / P38 / CORNER BRACE 2IN

WT-P38 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

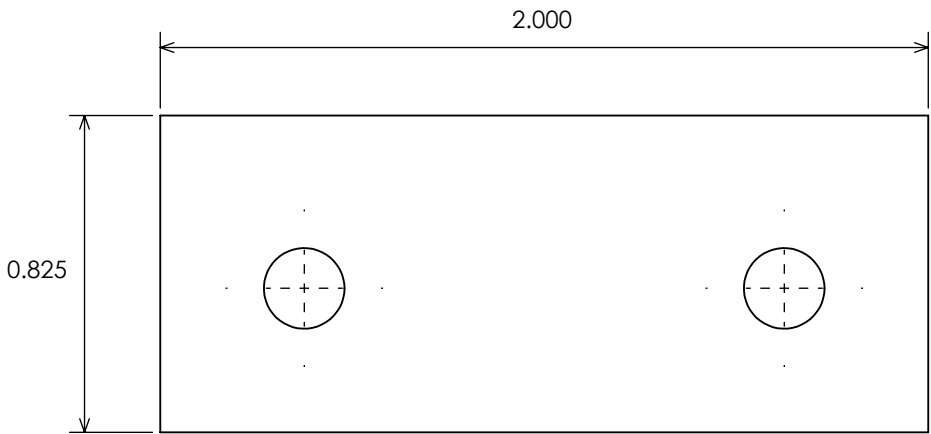
SIDE / X-Z | 2:1



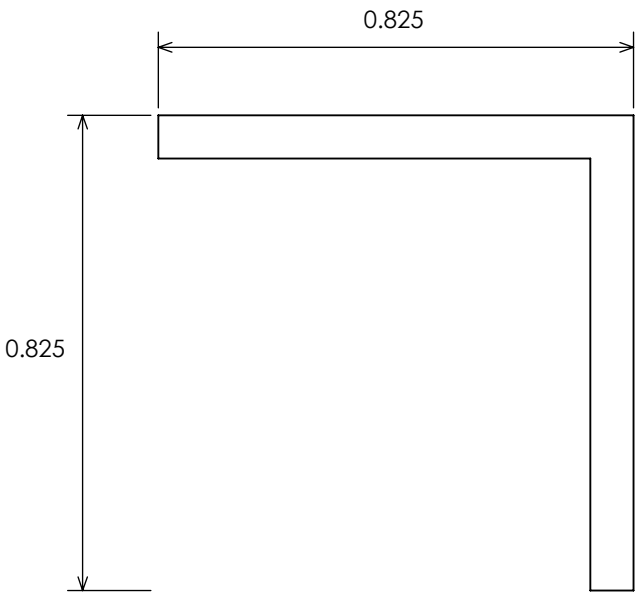
ISOMETRIC | 1.5:1



PLAN / X-Y | 2:1



END / Y-Z | 3:1



PART DIMENSIONS / DETAILS

Seam length X: 2.000
Flange leg length: 0.750 each
Nominal section gauge: 0.075
Inside bend angle: 90.000000 deg
4 x DIA 0.210 THRU
X hole stations from end:
0.375 / 1.625
Top flange hole center: Y = -0.375
Side flange: Z = -0.375000
Section profile in (u,v):
(-.750,0), (0,0), (0,-.750),
(.075,-.750), (.075,.075),
(-.750,.075). Sharp bend modeled.
Y = u; Z = v. Hole axes: Y and Z.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: Galvanized Steel

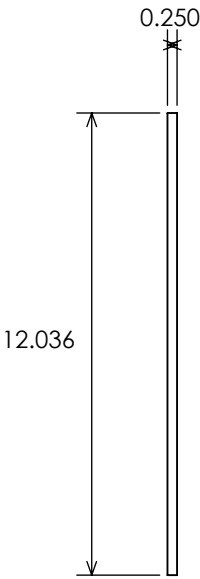
ASSEMBLY QTY: 8 | 10_Corner_Brace_2in.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

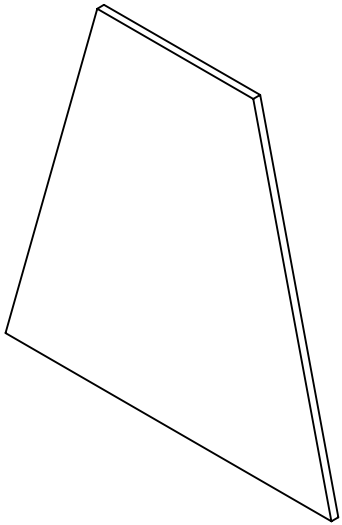
WIND TUNNEL / P39 / MEASURED TRAPEZOID A

WT-P39 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

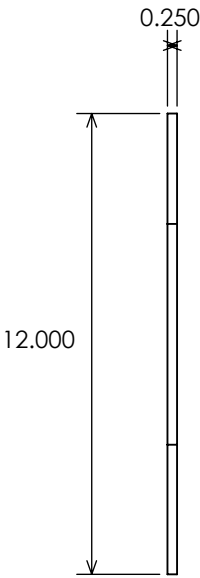
SIDE / X-Z | 1:5



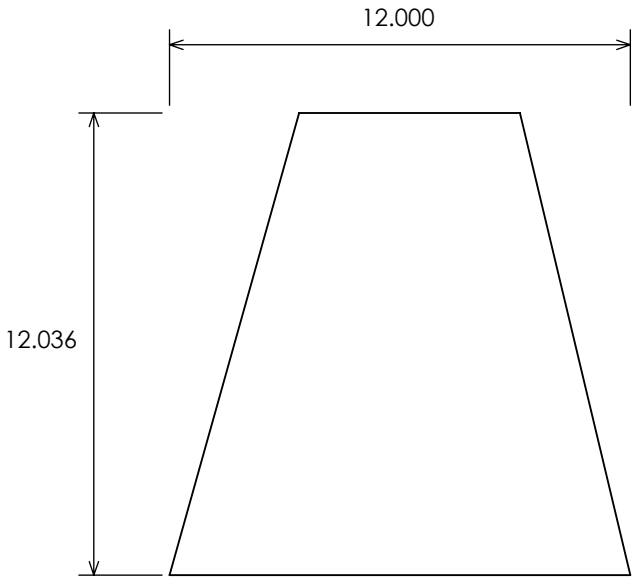
ISOMETRIC | 1:5



PLAN / X-Y | 1:5



END / Y-Z | 1:5



PART DIMENSIONS / DETAILS

Measured long edge: 12.000
Measured short edge: 5.750
Measured left edge: 12.500
Measured right edge: 12.375
Thickness X: 0.250 (assumed)
Profile height: 12.036104
Short-edge left offset: 3.373750
Profile coordinates (Y,Z):
A (0,0), B (12,0)
C (9.123750,12.036104)
D (3.373750,12.036104)
REFERENCE ONLY: not installed.
Source: photo 10_IMG_9661.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: BCX plywood - nominal

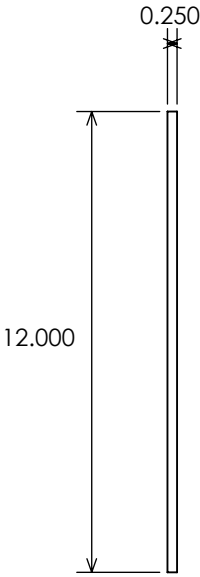
ASSEMBLY QTY: 0 | REF_Measured_Trapezoid_A.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

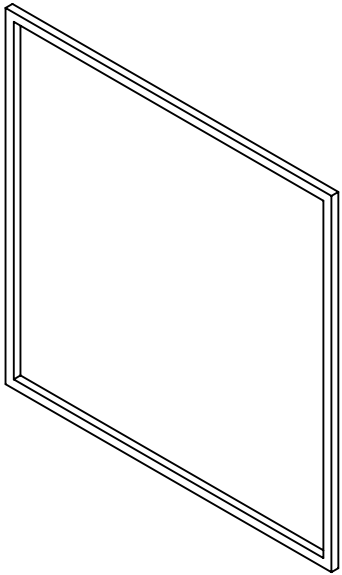
WIND TUNNEL / P40 / SCREEN RETAINING FRAME

WT-P40 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:5



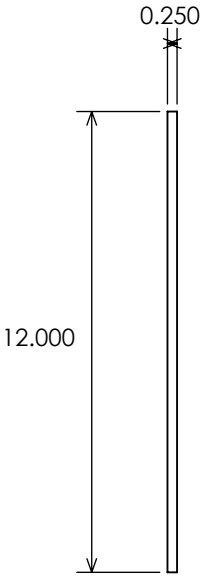
ISOMETRIC | 1:5



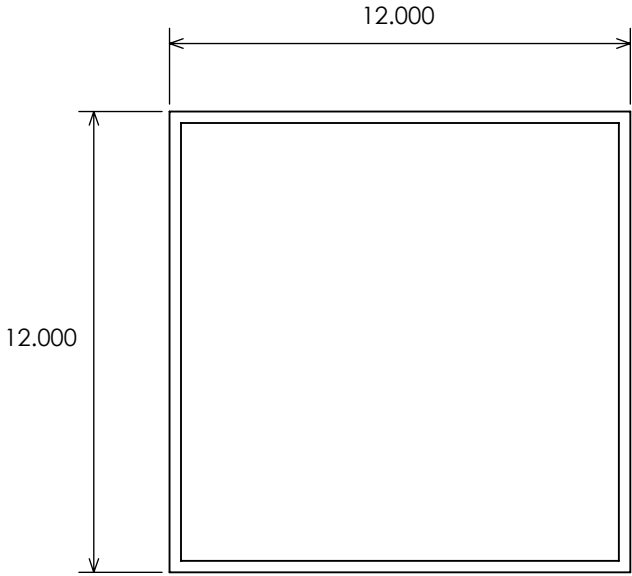
PART DIMENSIONS / DETAILS

Outer square: 12.000
Inner square: 11.400
Border width: 0.300
Axial thickness: 0.250
Assumed screen retaining frame.

PLAN / X-Y | 1:5



END / Y-Z | 1:5



MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: BCX plywood - nominal

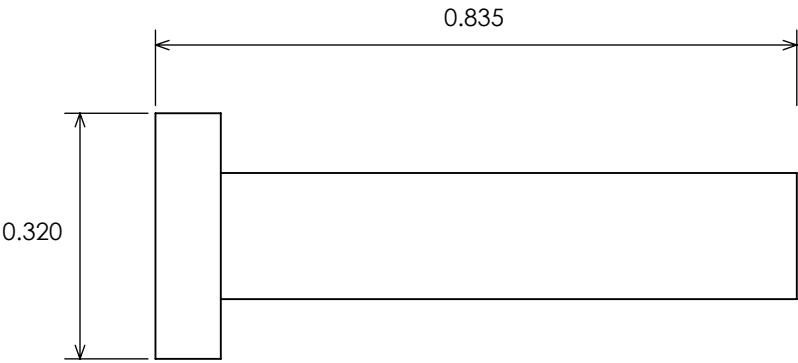
ASSEMBLY QTY: 2 | 11_Screen_Retaining_Frame.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

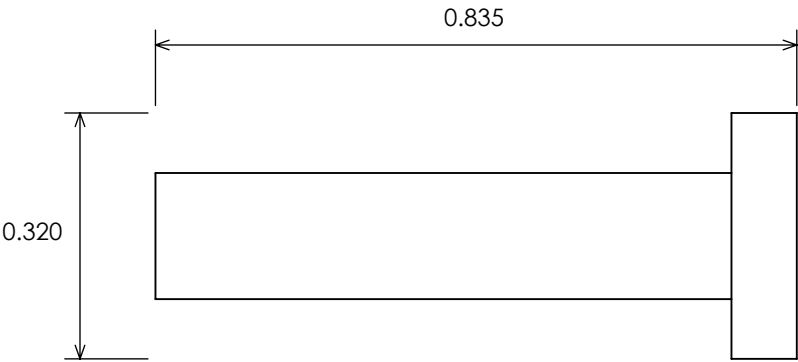
WIND TUNNEL / P41 / WOOD SCREW NO8 3 4IN

WT-P41 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

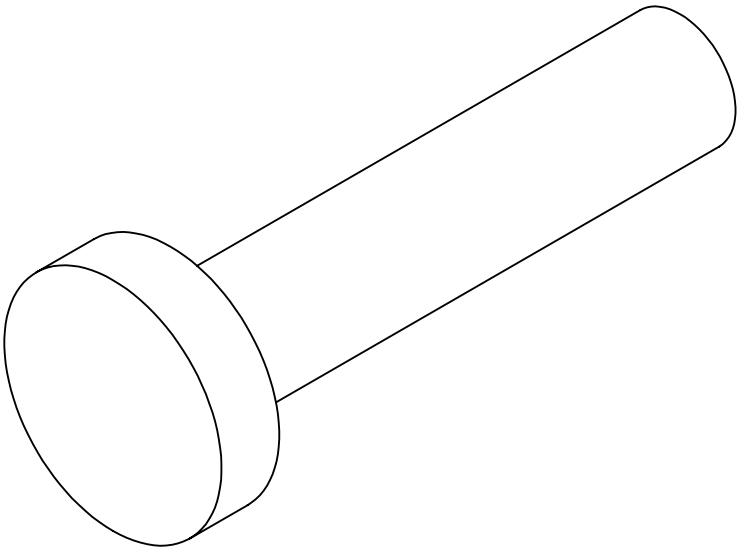
SIDE / X-Z | 4:1



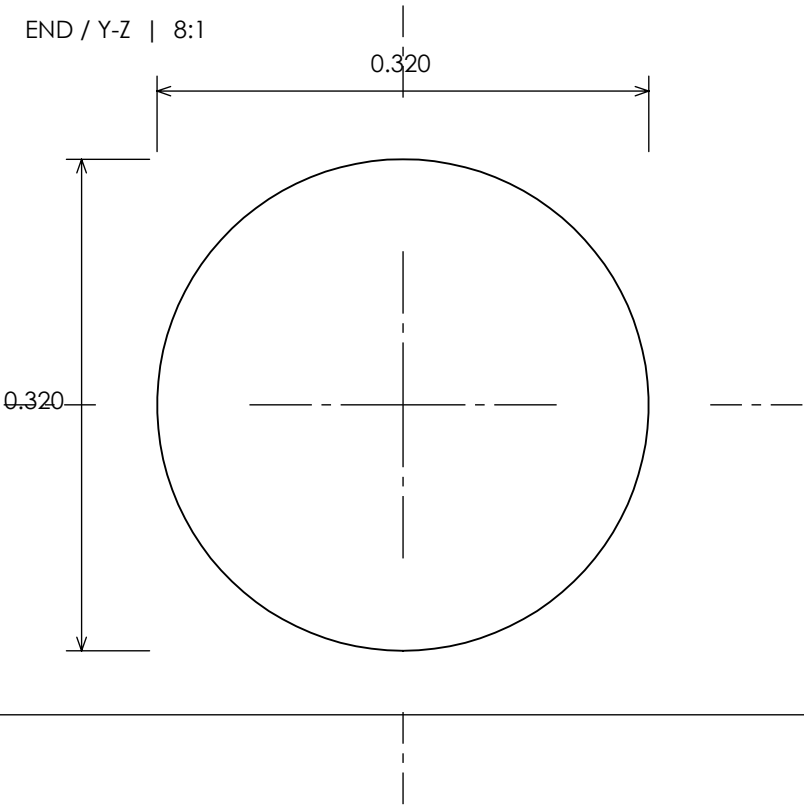
PLAN / X-Y | 4:1



ISOMETRIC | 5:1



END / Y-Z | 8:1



PART DIMENSIONS / DETAILS

- Shaft diameter: 4.17 mm (0.164173)
- Shaft length: 0.750
- Head diameter: 0.320
- Head axial thickness: 0.085
- Overall axial length: 0.835
- Nominal #8 x 3/4 wood screw.
- Smooth shank / round head proxy.
- Threads and countersink omitted.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: Galvanized Steel

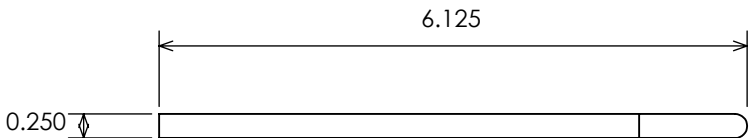
ASSEMBLY QTY: 128 | 11_Wood_Screw_No8_3_4in.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

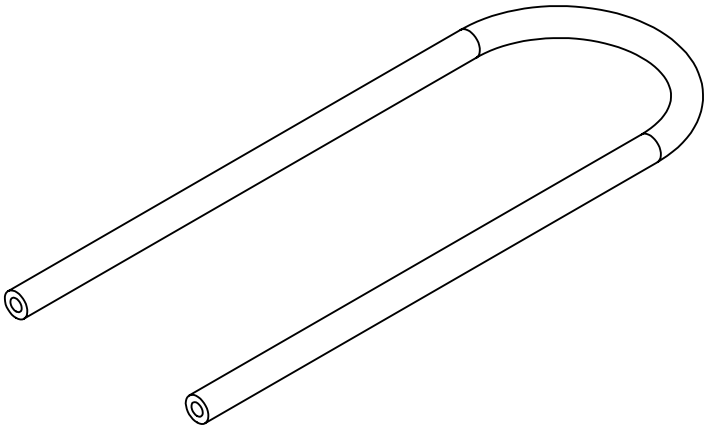
WIND TUNNEL / P42 / MANOMETER U TUBE

WT-P42 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

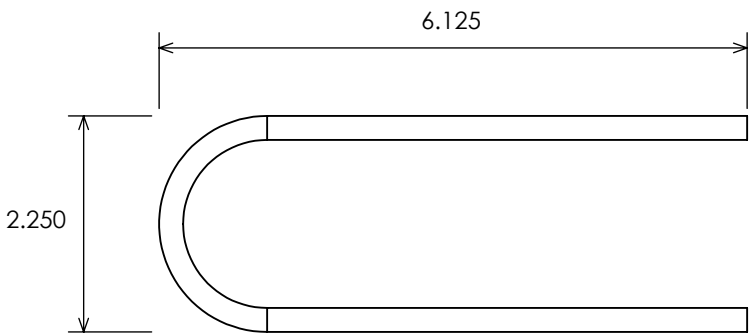
SIDE / X-Z | 1:2



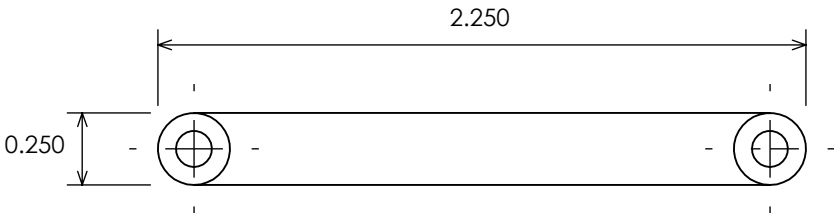
ISOMETRIC | 1:1.5



PLAN / X-Y | 1:2



END / Y-Z | 1.5:1



PART DIMENSIONS / DETAILS

- Tube outer diameter: 0.250
- Tube inner diameter: 0.125
- Wall thickness: 0.0625
- Bend centerline radius: 1.000
- Bend sweep: 180 deg
- Straight leg length: 5.000 each
- Leg center spacing: 2.000
- Tube cut length: 13.141593 REF
- Pressure taps / fluid omitted.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: Clear vinyl - assumed

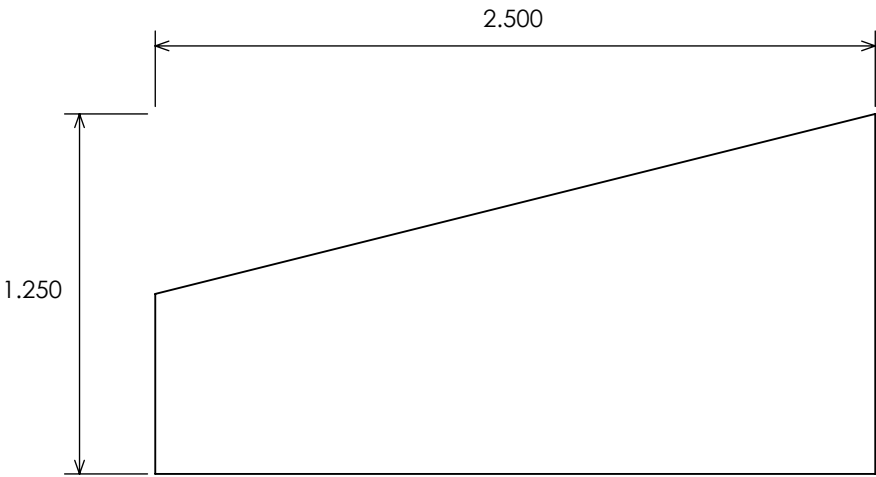
ASSEMBLY QTY: 1 | 11_Manometer_U_Tube.SLDPR

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

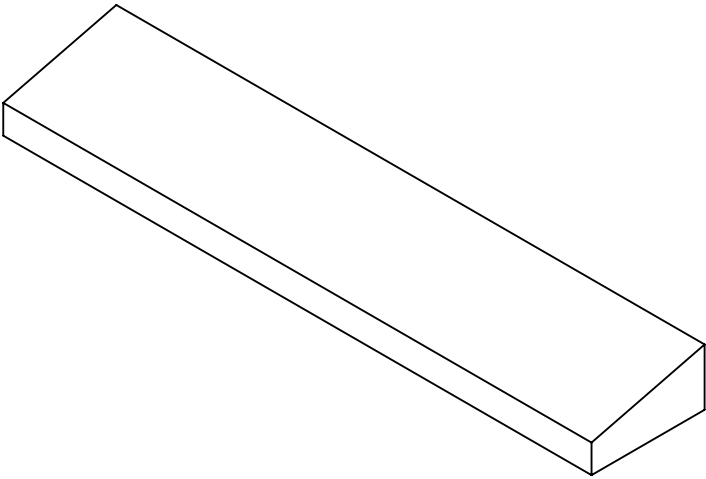
WIND TUNNEL / P43 / DIFFUSER SADDLE

WT-P43 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

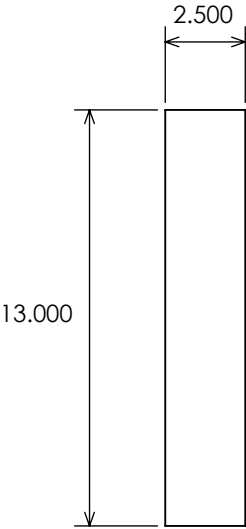
SIDE / X-Z | 1.5:1



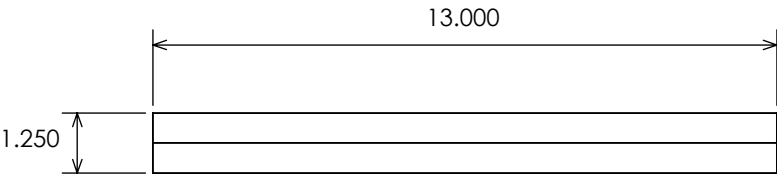
ISOMETRIC | 1:3



PLAN / X-Y | 1:6



END / Y-Z | 1:4



PART DIMENSIONS / DETAILS

- Axial depth X: 2.500
- Span Y: 13.000
- Entry height: 1.250
- Exit height: 0.625
- Top-face slope: 14.0362 deg
- Bottom is flat: Z = -6.250
- Top: Z = -5.000 to -5.625

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: Pine

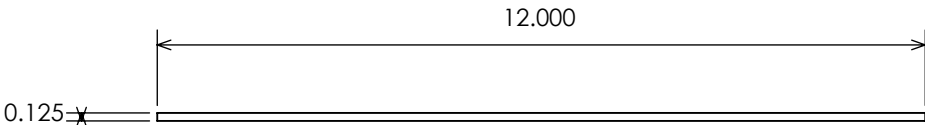
ASSEMBLY QTY: 1 | 11_Diffuser_Saddle.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

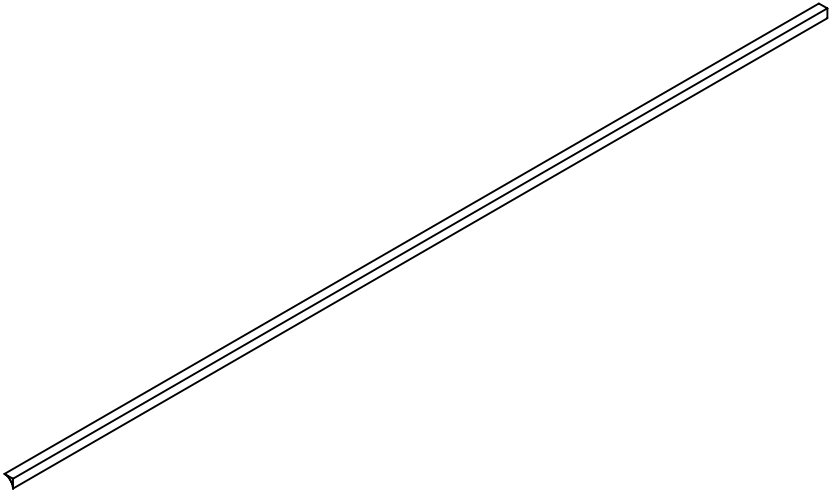
WIND TUNNEL / P44 / CAULK INLET CORNER

WT-P44 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

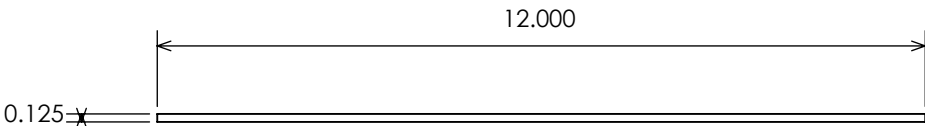
SIDE / X-Z | 1:3



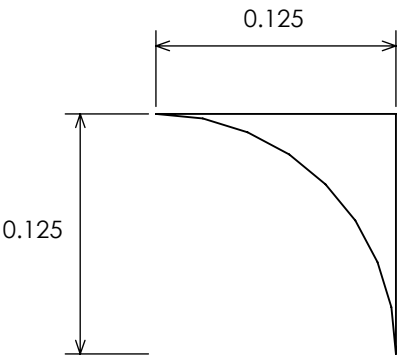
ISOMETRIC | 1:2



PLAN / X-Y | 1:3



END / Y-Z | 10:1



PART DIMENSIONS / DETAILS

Axial length X: 12.000
Fillet extent / radius: 0.1250
Start corner Y = Z = 6.000
End corner Y = Z = 6.000
Corner seam length: 12.000000
End profile: vertex (h,h), then
 $Y = h - r + r \cos(\alpha)$
 $Z = h - r + r \sin(\alpha)$
 $\alpha = 90, 78.75, \dots 0 \text{ deg}$
Eight straight facets approximate arc.
Straight loft between end profiles.
Sealant is an applied allowance;
exact as-built gap map unavailable.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: White caulk - assumed

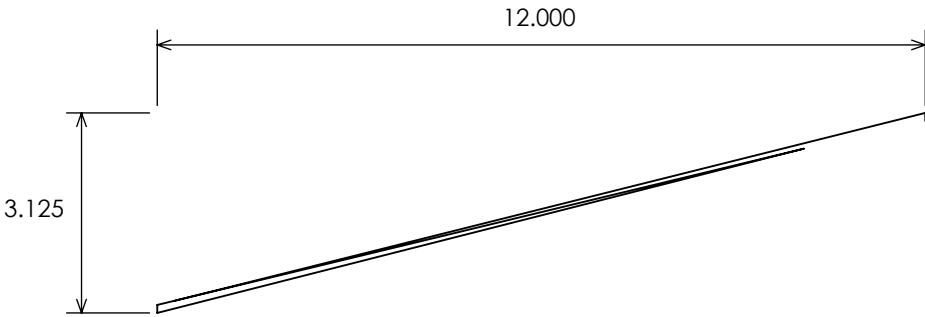
ASSEMBLY QTY: 4 | 12_Caulk_Inlet_Corner.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

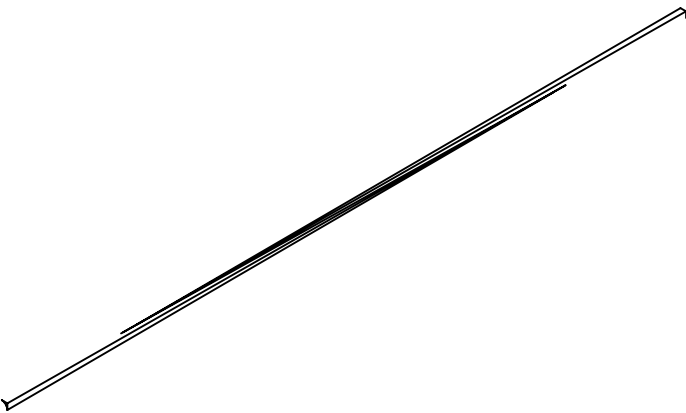
WIND TUNNEL / P45 / CAULK CONTRACTION CORNER

WT-P45 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

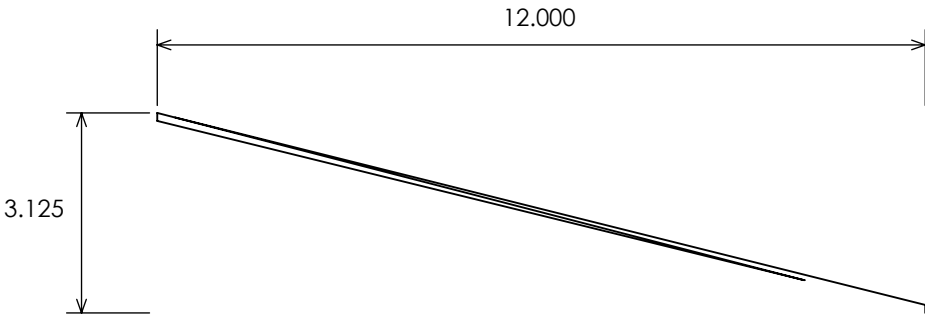
SIDE / X-Z | 1:3



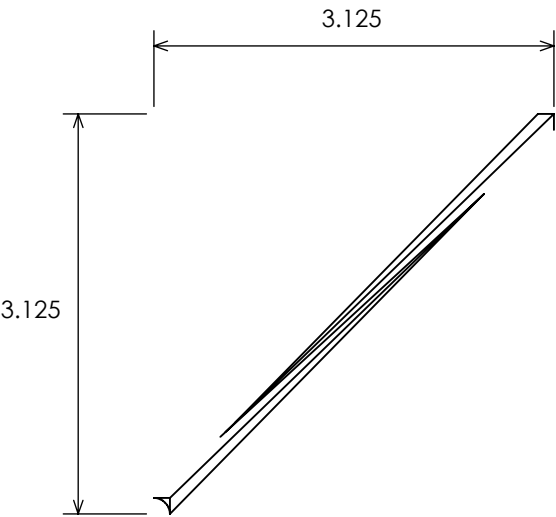
ISOMETRIC | 1:3



PLAN / X-Y | 1:3



END / Y-Z | 1:1.5



PART DIMENSIONS / DETAILS

Axial length X: 12.000
Fillet extent / radius: 0.1250
Start corner Y = Z = 6.000
End corner Y = Z = 3.000
Corner seam length: 12.727922
End profile: vertex (h,h), then
 $Y = h-r + r \cos(\alpha)$
 $Z = h-r + r \sin(\alpha)$
 $\alpha = 90, 78.75, \dots 0 \text{ deg}$
Eight straight facets approximate arc.
Straight loft between end profiles.
Sealant is an applied allowance;
exact as-built gap map unavailable.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: White caulk - assumed

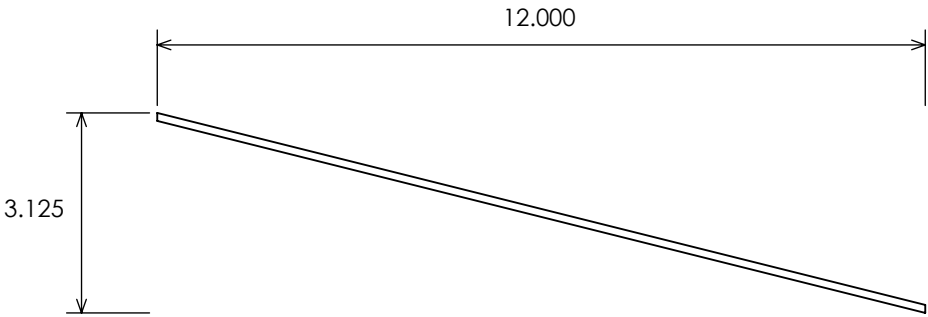
ASSEMBLY QTY: 4 | 12_Caulk_Contraction_Corner.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

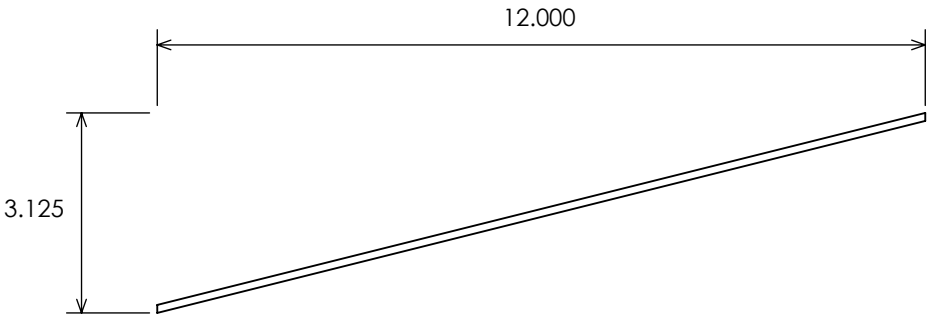
WIND TUNNEL / P46 / CAULK DIFFUSER CORNER

WT-P46 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

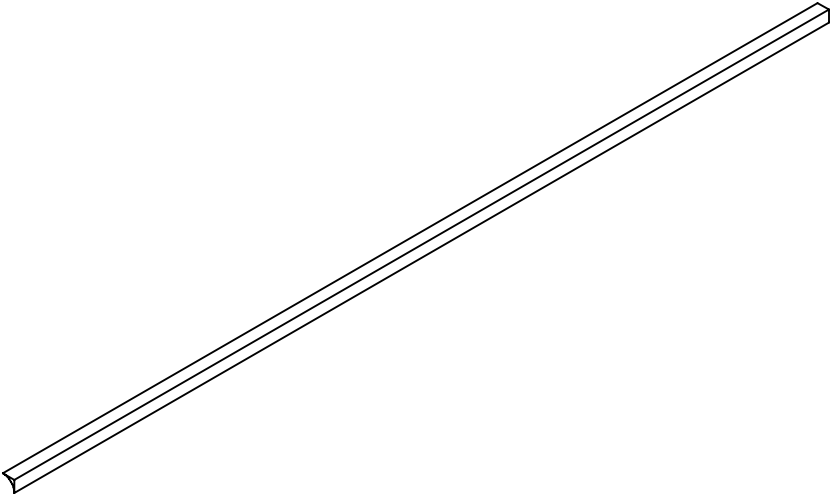
SIDE / X-Z | 1:3



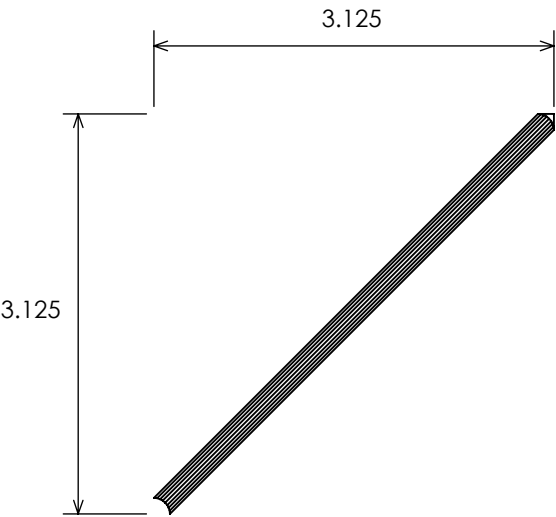
PLAN / X-Y | 1:3



ISOMETRIC | 1:1.5



END / Y-Z | 1:1.5



PART DIMENSIONS / DETAILS

Axial length X: 12.000
Fillet extent / radius: 0.1250
Start corner Y = Z = 3.000
End corner Y = Z = 6.000
Corner seam length: 12.727922
End profile: vertex (h,h), then
 $Y = h-r + r \cos(\alpha)$
 $Z = h-r + r \sin(\alpha)$
 $\alpha = 90, 78.75, \dots 0 \text{ deg}$
Eight straight facets approximate arc.
Straight loft between end profiles.
Sealant is an applied allowance;
exact as-built gap map unavailable.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: White caulk - assumed

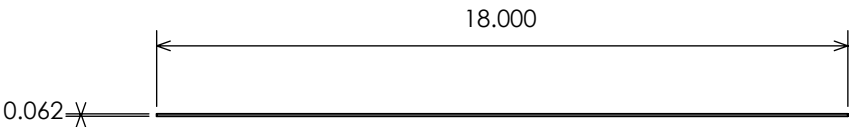
ASSEMBLY QTY: 4 | 12_Caulk_Diffuser_Corner.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

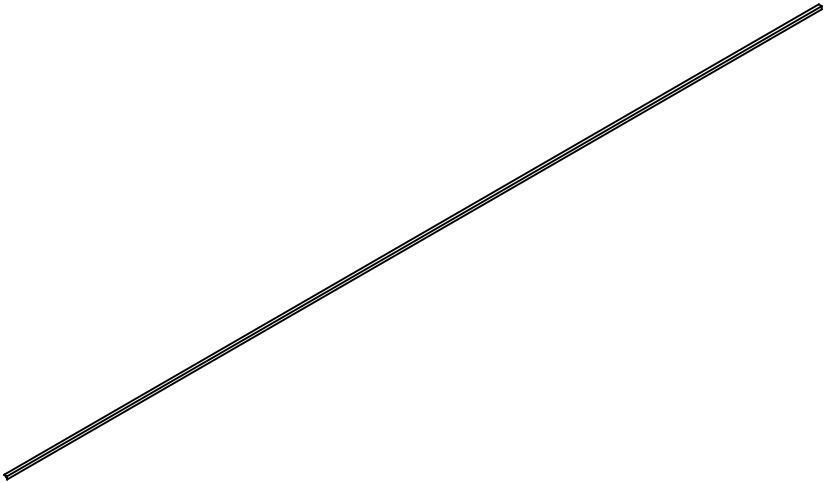
WIND TUNNEL / P47 / CAULK ACRYLIC FLOOR SEAM

WT-P47 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

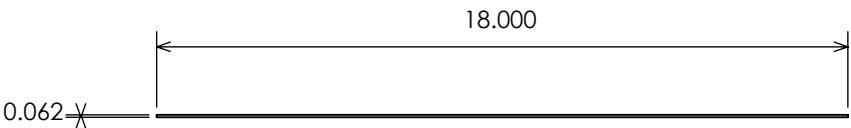
SIDE / X-Z | 1:5



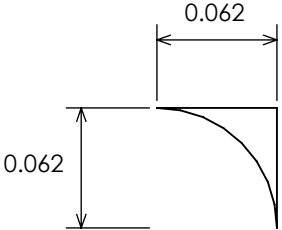
ISOMETRIC | 1:3



PLAN / X-Y | 1:5



END / Y-Z | 10:1



PART DIMENSIONS / DETAILS

Axial length X: 18.000
Fillet extent / radius: 0.0625
Start corner Y = Z = 3.000
End corner Y = Z = 3.000
Corner seam length: 18.000000
End profile: vertex (h,h), then
 $Y = h - r + r \cos(\alpha)$
 $Z = h - r + r \sin(\alpha)$
 $\alpha = 90, 78.75, \dots 0 \text{ deg}$
Eight straight facets approximate arc.
Straight loft between end profiles.
Sealant is an applied allowance;
exact as-built gap map unavailable.

MODEL AXES

X: extrusion / axial direction
Y: transverse; Z: vertical
Views independently scaled as labeled.
Overall dimensions rounded to 0.001 in.
Detail values control small features.

MATERIAL: White caulk - assumed

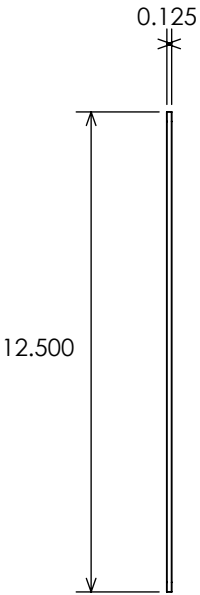
ASSEMBLY QTY: 2 | 12_Caulk_Acrylic_Floor_Seam.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

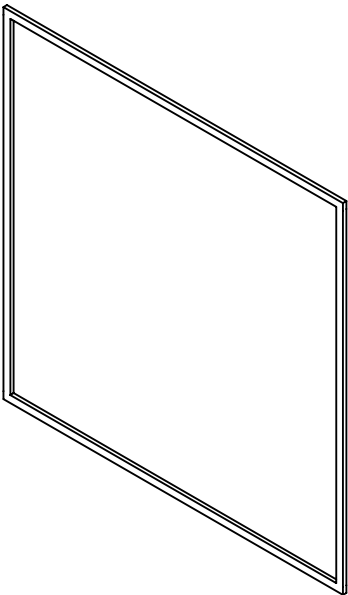
WIND TUNNEL / P48 / CAULK INLET CONTRACTION JOINT

WT-P48 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

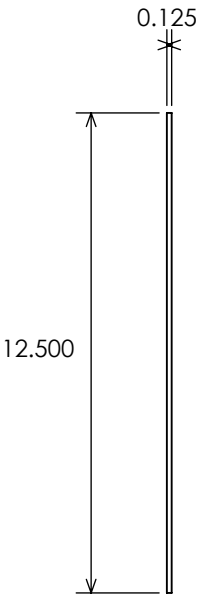
SIDE / X-Z | 1:5



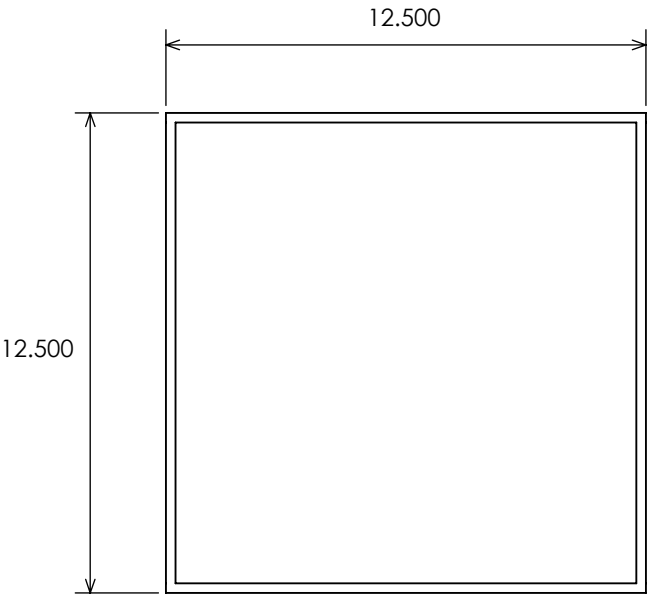
ISOMETRIC | 1:5



PLAN / X-Y | 1:5



END / Y-Z | 1:5



PART DIMENSIONS / DETAILS

- Axial gap X: 0.125
- Clear bore each end: 12.000 square
- Entry radial wall: 0.250
- Exit radial wall: 0.250
- Entry outer square: 12.500
- Exit outer square: 12.500
- Entry Y center: 0.000
- Exit Y center: 0.000
- Z center: 0.000 at both ends
- Four wall lofts; open airflow bore.
- CAD solid bodies: 1
- Gap and transverse shift assumed.

MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: White caulk - assumed

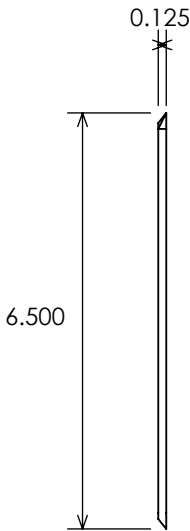
ASSEMBLY QTY: 1 | 12_Caulk_Inlet_Contraction_Joint.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

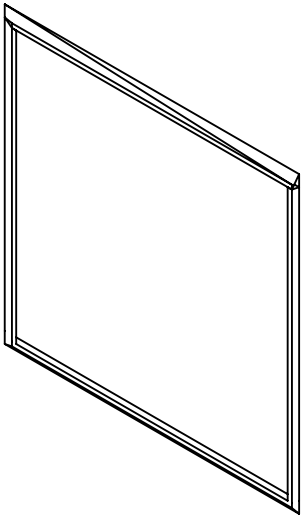
WIND TUNNEL / P49 / CAULK CONTRACTION ACRYLIC JOINT

WT-P49 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:3



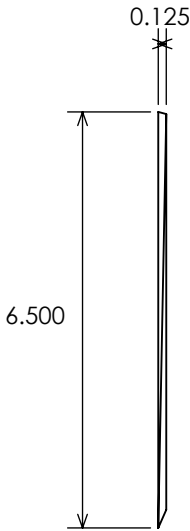
ISOMETRIC | 1:3



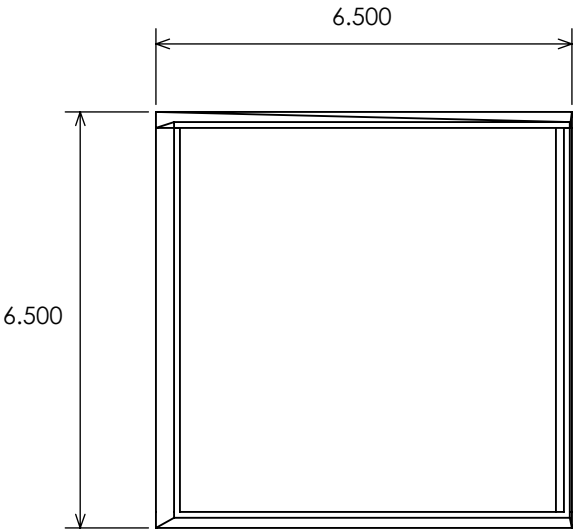
PART DIMENSIONS / DETAILS

- Axial gap X: 0.125
- Clear bore each end: 6.000 square
- Entry radial wall: 0.250
- Exit radial wall: 0.093
- Entry outer square: 6.500
- Exit outer square: 6.186
- Entry Y center: 0.000
- Exit Y center: 0.125
- Z center: 0.000 at both ends
- Four wall lofts; open airflow bore.
- CAD solid bodies: 2
- Gap and transverse shift assumed.

PLAN / X-Y | 1:3



END / Y-Z | 1:3



MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: White caulk - assumed

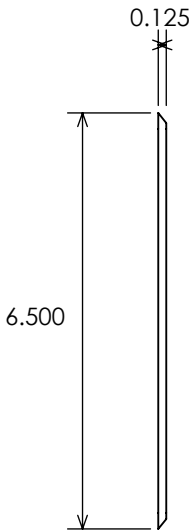
ASSEMBLY QTY: 1 | 12_Caulk_Contraction_Acrylic_Joint.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.

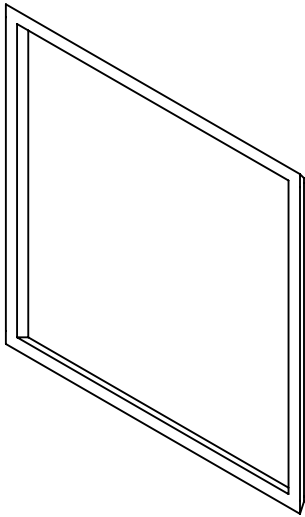
WIND TUNNEL / P50 / CAULK ACRYLIC DIFFUSER JOINT

WT-P50 REV D 2026-10-01 | NOMINAL CAD GEOMETRY | INCHES UNLESS MARKED mm

SIDE / X-Z | 1:3



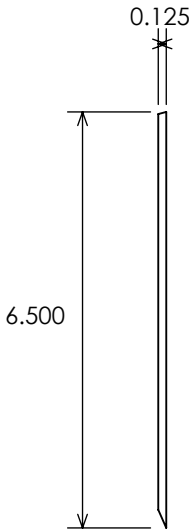
ISOMETRIC | 1:3



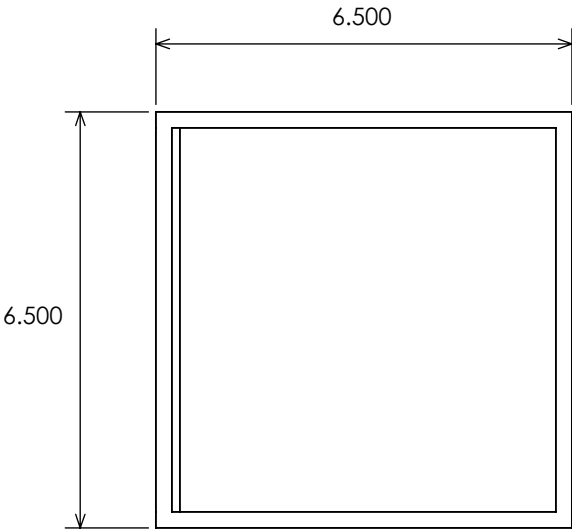
PART DIMENSIONS / DETAILS

- Axial gap X: 0.125
- Clear bore each end: 6.000 square
- Entry radial wall: 0.093
- Exit radial wall: 0.250
- Entry outer square: 6.186
- Exit outer square: 6.500
- Entry Y center: 0.125
- Exit Y center: 0.000
- Z center: 0.000 at both ends
- Four wall lofts; open airflow bore.
- CAD solid bodies: 1
- Gap and transverse shift assumed.

PLAN / X-Y | 1:3



END / Y-Z | 1:3



MODEL AXES

- X: extrusion / axial direction
- Y: transverse; Z: vertical
- Views independently scaled as labeled.
- Overall dimensions rounded to 0.001 in.
- Detail values control small features.

MATERIAL: White caulk - assumed

ASSEMBLY QTY: 1 | 12_Caulk_Acrylic_Diffuser_Joint.SLDPRT

Model-linked views. Dimensions describe nominal CAD; verify assumptions against the physical build.
No unmeasured holes, threads, fasteners or purchased-component internals have been inferred beyond the modeled geometry.